Information in Sequential Decision Making with Rejection Externalities *

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February 22, 2023

Abstract

Applicants of unknown types apply to multiple evaluators, seeking funding from one. Evaluators only want to fund High types, and see costless and private signals of applicants' (i) types and (ii) rejection histories. A candidate rejected by an evaluator applies to other evaluators, concocting a *rejection externality*: evaluators' decisions affect each other's applicant pools. I study how evaluators' equilibrium payoffs respond to (Blackwell) improvements in either signal. First, I characterise the effect of improving the type signal when the history signal is uninformative. This hinges on whether (i) the improvement is through stronger good news or bad news, (ii) the rejection externality is present. I describe the decision rules maximising evaluators' total payoffs, and the effect of improving type signals when these are imposed. Then, I show that a fully informative history signal maximises evaluators' total payoffs irrespective of the type signal. Nonetheless, some Blackwell improvements of evaluators' history signals might reduce equilibrium payoffs.

1 Introduction

Consider a setting where one must apply for an award to be considered for it, but can apply to many evaluators until she eventually succeeds or loses hope. A manuscript author seeking a prestigious publication can submit to multiple reputable journals before considering less prestigious outlets. Likewise, an entrepreneur can pitch his business idea to several prospective investors before discarding it.

In such settings, a *rejection externality* complicates evaluators' selection problem. Some applications an evaluator receives will have come from candidates who were previously rejected by other evaluators. In turn, some applicants she rejects will apply to her peers later. Hence, an evaluator's approval decisions will shape others' applicant pools. Continuing our examples, a journal editor will be aware that manuscripts she receives might have been rejected from other journals due to flaws her evaluation missed. Similarly, an investor considering whether to

*I thank Ian Jewitt, Margaret Meyer, Daniel Quigley, Paula Onuchic and Ludvig Sinander for long discussions and invaluable guidance. I also thank Alex Teytelboym, Ines Moreno de Barreda, Peter Eső, Manos Perdikakis, Inbar Amit, Michael Eldar, Alae Baha, seminar audiences at the Oxford Student Micro Theory Workshop, and speakers at the Nuffield Economic Theory Seminar for their feedback and comments.

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fund an entrepreneur will wonder if others turned the entrepreneur down before, following an unfavourable verdict on the project's promise.

Information an evaluator has on her applicant's *quality* and *past rejections* will guide her approval decisions. In such environments, endowing evaluators with more information on either front will have a twofold effect on their payoffs. It will have a favourable *direct effect*: evaluators' ability to tailor their decisions against uncertainty will increase. This direct effect has been extensively studied, particularly in the context of single person decision problems[1] In addition, an *indirect effect* will emerge through the rejection externality. An evaluator's applicant pool is partly determined by others' rejection decisions. Evaluators' information on their applicants influences the composition of their applicant pool, as it guides these decisions. Hence, improving evaluators' information affects both their ability to respond to uncertainty, and *the uncertainty they face itself.*

This paper studies how (Blackwell) improvements to evaluators' private information influences their payoffs through the direct and indirect effects, in these settings. To this end, I build a model I call the *K-evaluation game.* The K-evaluation game unfolds over discrete time. Every period, a mass of *candidates* are born, each with a *High*, or *Low* type. A candidate's type is unknown to all players in the game, including himself [2] He sequentially applies to $K \geq 2$ symmetric *evaluators* for approval, in a random order. A candidate applies to an evaluator at most once, and retires either when he receives an approval, or has applied to every evaluator.

An evaluator wants to approve only *High* types. Her applicants consist of candidates born that period as well as those rejected by other evaluators earlier. For every applicant, she observes a (binary) *type signal* and a *history signal* to guide her decisions. While the type signal is informative about an applicant's type, the latter signals the number of his prior rejections (age). I focus on the symmetric, steady state, Bayes Nash Equilibrium of a K-evaluation game, which I call the S-EQ.

Special cases of the *K-evaluation game* have received notable attention in economics. The *public* K-evaluation game, where the history signal is *fully informative*, mirrors the canonical social learning model in Bikhchandani et al., 1992 (BHW). Its only difference from BHW is that the public K-evaluation game is only prone to *rejection cascades* (in BHW's language); as approved candidates retire.

On the opposite end of the spectrum, there is the *private* K-evaluation game, where the history signal is fully *uninformative.* I show that that the private K-evaluation game is isomorphic to a *simultaneous* K-evaluation game. In the latter, candidates apply *simultaneously to all evaluators* upon birth, and accept one randomly if multiple evaluators offer an approval.

I first ask how how evaluators' S-EQ payoffs in the *private K-evaluation game* changes when their type signal is made (Blackwell) more informative. To answer it, I develop a language describing the two forms in which a binary signal can be improved. I say the new signal delivers *stronger good* (*bad*) *news* when the high (low) signal realisation is stronger evidence for the *High* (*Low*) type. A binary signal is (Blackwell) improved either when it delivers stronger good news, and/or stronger bad news. Additionally, I say *history is irrelevant* if a candidate's application history is irrelevant for an evaluator's decision conditional on the type signal. This happens if

[1] Especially following Blackwell and Girshick, 1954's seminal contribution.

[2] The analysis is identical if candidates are assumed to know their types.

it is optimal to approve a candidate upon a high type signal even if all but one evaluators saw low type signals before. Otherwise, I say history is relevant.

I show that the answer to this first question is determined entirely by (i) the form of the Blackwell improvement and (ii) whether *history is relevant*. I find that S-EQ payoffs weakly increase with *stronger good news*. If history is irrelevant, they weakly increase with *stronger bad news*, too. These mirror their effect in single person decision problems. However, whenever history is relevant, S-EQ payoffs weakly decrease with *stronger bad news*. This is in stark contrast to single person decision problems, where more information always benefits the decision maker.

This contrast owes to the absence of the indirect effect in single person decision problems. Although the direct effect of stronger bad news is always positive[3], I show that its indirect effect is always weakly positive. Put simply, despite helping evaluators tailor their actions better to uncertainty, a type signal delivering stronger bad news always exacerbates adverse selection into the applicant pool. My first main result establishes that this negative indirect effect always outweighs the positive indirect effect, unless history is irrelevant.

I continue with a focus on the private K-evaluation game, and characterise the strategy profile which maximises the sum of evaluators' payoffs. I first find the optimal *symmetric* strategy profile. As in the S-EQ, evaluators always reject upon a low signal under this strategy profile. However, their approval probability upon a high signal is strictly lower than it is in the S-EQ, unless history is irrelevant. This helps replenish the applicant pool with favourably evaluated applicants. As such, it alleviates adverse selection into the applicant pool; or, mutes the indirect effect, raising evaluators' payoffs.

Under this strategy profile, improving the type signal always weakly increases payoffs. However, stronger bad news never strictly increase payoffs, unless history is irrelevant. This is best understood through Blackwell's Theorem. Blackwell's Theorem clarifies that for a given type signal, an outcome is supported by some decision rule if and only if it can be supported by some garbling of the type signal. Our first main result established that stronger bad news decrease S-EQ payoffs whenever history is relevant. Hence, the garbling maximising evaluators' payoffs reduces the strength of bad news until history is irrelevant. The optimal symmetric strategy profile induces the outcome this optimal garbling would. When history is relevant, any type signal delivering stronger bad news leaves this garbling unchanged. Hence, it does not increase evaluators' payoffs under this strategy profile.

I then turn to the *asymmetric* strategy profile maximising the sum of evaluators' payoffs. This optimal strategy profile is remarkably simple. It leaves only a subset of evaluators *active*. Active evaluators always approve upon seeing a high signal. The remaining, *inactive* evaluators never approve applicants. The number of active evaluators is the maximum number of evaluators with whom history would be irrelevant.

Next, I observe that the outcome generated by the optimal asymmetric strategy profile mirrors the S-EQ of the corresponding *public* K-evaluation game. In this equilibrium, a candidate is approved by the initial evaluators he applies to, provided they see a high signal. Hence, their rejection reveals a low signal. Once the low signals the candidate accumulates cannot be overturned by a single high signal, a rejection cascade a lá BHW emerges. All subsequent evaluators reject the candidate indiscriminately. The optimal number of active evaluators under

[3] As is well known from Blackwell's Theorem

the optimal symmetric strategy profile equals the number of rejections a candidate can receive before a rejection cascade begins.

This outcome equivalence clarifies that regardless of the type signal, a fully informative history signal yields higher S-EQ payoffs than an uninformative history signal. I show that a stronger result is true: a fully informative history signal yields higher S-EQ payoffs than *any* history signal, irrespective of the type signal. However, the Blackwell improvement of an arbitrary history signal does not necessarily increase S-EQ payoffs. I conclude by constructing a K-evaluation game where S-EQ payoffs can be increased by coarsening a history signal.

1.1 Related Literature

My model is most closely related to the observational learning literature. In observational learning models, a sequence of individuals decide whether to *adopt* or *reject* some behaviour. An individual's payoff depends on her decision *and* an unknown state. Every individual receives private information about the state before their decision. Individuals observe their predecessors' decisions as well, also informative about the unknown state.

The canonical observational learning models were developed by Bikhchandani et al., 1992 (BHW) and Banerjee, 1992 to explain the phenomena of *herding* or *informational cascading*. My model inherits key properties of the BHW model. However, it has two key departures from their structure.

The first is in how early decision makers' choices affect whether the decision will be presented to later decision makers. In BHW, every individual in the sequence is eventually presented with the decision. In the K-evaluation game, once an evaluator approves an applicant, the applicant retires. The remaining evaluators can no longer consider him. This is critical in my model, as it generates the fundamental economic force of interest, the *rejection externality*.

My second departure is in what decision makers observe about their predecessors' actions. In BHW, individuals see *all* their predecessors' choices. The K-evaluation game allows evaluators' information about the number of their predecessors who evaluated and rejected their applicant earlier to be arbitrary.

Both Lockwood, 1991 and Herrera and Hörner, 2013 share the first departure. They also assume that decision makers see noisy, though not arbitrary, information about their predecessors' decisions. In Lockwood's labour market hiring model predating BHW and Banerjee, 1992, a candidate sequentially applies to vacancies which arrive at a Poisson rate. The candidate stops applying once he makes a successful application. Upon an application, prospective employers see the applicant's unemployment duration, but not how many times he got turned down earlier.

In Herrera and Hörner, 2013, agents sequentially consider whether or not to invest in a project. Agents' arrivals are random, and is governed by a Poisson process. Upon her arrival, an agent receives private information about the project's likelihood to succeed. She alsos see how much time elapsed until her arrival, as well as the number of her predecessors who invested. She, nonetheless, does not see how many of her predecessors considered but refused to invest. In one variant of their model, the project disappears once one agent invests.

In both models, the decision maker worries that her predecessors might have received unfavourable information about the unknown state. Improving decision makers' private informa-

tion might exacerbate this worry, complicating the effect this has on their payoffs. Nonetheless, this complication is outside both models' focus, while the trade-off it stems from lies in the heart of my model. In the private K-evaluation game, I fully characterise how evaluators' payoffs respond to different *kinds* of improvements in their private information. I find that the rejection externality causes results to depart substantially from the single person decision making benchmark.

I also note that the answer to this question depends critically on what information evaluators see about their predecessors' decisions. While strengthening bad news lowers evaluators' payoffs in the private K-evaluation game, any improvement in information increases evaluators' payoffs in the public K-evaluation game. The models above are not conducive to this observation, as they fix the information a decision maker sees about her predecessors' decisions.

Sgroi, 2002, Parakhonyak and Vikander, 2023, and Acemoglu et al., 2022 also look at observational learning environments, and consider how a designer might benefit from coarsening decision makers' information on their predecessors actions. Sgroi considers a firm selling its product to consumers sequentially, a lá BHW. He studies how it might benefit from forcing an initial subset of consumers – guinea pigs – to a decision without any predecessors. This ensures that guinea pigs' private information is fully inferred by the remaining customers, with no risk of herding.

Parakhonyak and Vikander, 2023 study a two period model where a firm who receives a batch of customers each period. Customers decide whether to purchase given their private information on the quality of the product. However, customers who arrive in the second period also observe the decisions of those in the first batch. They find that the firm can benefit from thwarting first period supply, which obscures information about the true demand it faced in this period.

Both papers observe that coarsening information about history can increase a designers' payoff. In my model, I show that a fully informative history signal is best for evaluators' equilibrium payoffs. Nonetheless, coarsening a particular history signal might be beneficial for them. The intuition and economic mechanism underlying this result differs from both of these models: coarser history signals alleviate the rejection externality evaluators face by changing their strategies.

Acemoglu et al., 2022 consider an infinite horizon observational learning setting, a lá Smith and Sørensen, 2000. They ask whether a sales platform might increase the *speed* of learning by coarsening buyers' information about past consumers' feedback, and provide a positive answer. This operates through changing consumers' decisions to participate in this market, and their equilibrium inferences from the information they receive. They do not consider how changing decision makers' private information interacts with their information on their predecessors' feedback.

2 The Model

The *K-evaluation game* unfolds over doubly infinite[4] discrete time. Every period, a unit mass of *candidates* (he) are born. Upon birth, candidate i draws a type $\theta_i \in \{H, L\}$; for *High* and

[4]There is no beginning, nor an end to the time horizon.

Low respectively. Draws of θ_i are independent across candidates:

$$\mathbb{P}(\theta = H) = p_0 \in (0,1)$$

where p_0 is common knowledge. I call p_0 the *fresh prior*. A candidate born in period T also draws an index $\iota \sim U[T, T+1]$. In period $t \geq T$, this candidate's *age* is given by $\tau := t - \lfloor \iota \rfloor$ [5].

Candidates sequentially apply to $K \geq 2$ infinitely lived *evaluators* (she) for approval. These evaluators are labelled as $j \in \{1, 2, ..., K\}$. A candidate *retires* once he is approved by an evaluator, or is rejected by all K evaluators; i.e. $\tau = K$. Each period preceding retirement, he picks one evaluator he has not applied to yet, uniformly at random, and applies to her for approval.

For each applicant she gets, an evaluator observes a realization of the *type signal* X. X is a binary random variable. Conditional on θ, it is independent of the applicant's age τ, and IID across evaluators with type dependent CDF F^θ:

$$X \mid \theta \overset{\text{IID}}{\sim} F^\theta \qquad\qquad X \mid \theta \perp\!\!\!\perp \tau$$

The type signal X is described by the possible values it can take, $\{x_f, x_s\} \subset (0,1)$. I name these realisations of the type signal after the *likelihood ratios* attached to them:

$$\frac{F^H(x_f)}{F^L(x_f)} = \frac{x_f}{1-x_f} \leq 1 \qquad\qquad \frac{1-F^H(x_f)}{1-F^L(x_f)} = \frac{x_s}{1-x_s} \geq 1 \tag{2.1}$$

Neither of these signal realisations are fully revealing since $0 < x_f < 0.5 < x_s < 1$. Note that 2.1 implies that a Bayesian statistician who initially believes each state is equally likely assigns probability x_f (x_s) to the state $\theta = H$ upon observing the signal realisation x_f (x_s). Where convenient, I call the realisation $x = x_f$ a *Fail*, and $x = x_s$ a *Success*. Also for convenience, I occasionally refer to this type signal X as (x_f, x_s).

I denote the conditional probabilities associated with the type signal X as $\mu_0(x_f, x_s)$ and $\mu_1(x_f, x_s)$:

$$\mu_0(x_f, x_s) := F^L(x_f) \qquad\qquad \mu_1(x_f, x_s) := 1 - F^H(x_f) \tag{2.2}$$

μ_0 (μ_1) is the probability a *Low* (*High*) type *Fails* (*Succeeds*) under X. Equalities 2.1 and 2.2 define a system of two linear equations which can be succinctly summarized as:

$$(1-x_s) \times \mu_1 = x_s \times (1-\mu_0) \tag{2.3}$$

$$(1-x_f) \times (1-\mu_1) = x_f \times \mu_0 \tag{2.4}$$

This linearly independent system of two equations in two unknowns determines (μ_0, μ_1) uniquely.

I focus on binary type signals I call *individually obedient* (or, which satisfy the *individual obedience constraints*) given the prior $p_0 := \mathbb{P}(\theta = H)$ and funding cost $c \in (0,1)$. This simply means that if there were a single evaluator with subjective prior p_0 and cost c, she would find it optimal to follow the recommendation of his private type signal.

[5] $f(x) = \lfloor x \rfloor$ is the integer floor function.

Definition 2.1. The type signal (x_f, x_s) is *individually obedient* given the fresh prior p_0 if the inequalities:

$$\frac{p_0}{1-p_0}\frac{x_s}{1-x_s} \geq \frac{c}{1-c} \tag{O-S}$$

$$\frac{p_0}{1-p_0}\frac{x_f}{1-x_f} \leq \frac{c}{1-c} \tag{O-F}$$

hold. The set of such binary experiments is denoted $\Omega^{p_0,c}$. In other words:

$$\Omega^{p_0,c} = (0, B_f] \times [B_s, 1) \subset \mathbb{R}^2$$

where:

$$B_f = \min\left\{0.5, \frac{c(1-p_0)}{c(1-p_0)+(1-c)p_0}\right\} \qquad B_s = \max\left\{0.5, \frac{c(1-p_0)}{c(1-p_0)+(1-c)p_0}\right\}$$

If both O-S and O-F hold with equality, we get $x_f = x_s = 0.5$. This implies that X is uninformative. I focus exclusively on *informative* type signals:

Definition 2.2. The type signal X is informative if $x_s - x_f \neq 0$.

Like an applicant's type θ, his age τ, which reveals the number of previous rejections he receives, remains *hidden* to the evaluator he applies to. However, prior to her decision, the evaluator observes the realisation of the *history signal* Z for her applicant. Z reveals some information about τ to the evaluator. Conditional on τ, Z is IID across evaluators, independent from θ, and follows distribution G^τ:

$$Z \mid \tau \overset{\text{IID}}{\sim} G^\tau \qquad\qquad Z \mid \tau \perp\!\!\!\perp \theta$$

The realisations of both X and Z are costless and private to evaluators.

When experiment $X^\dagger$ is *Blackwell more informative than* (or simply *more informative*) X, I write $X^\dagger \succ X$ [6]. When such, I say $X^\dagger$ *(Blackwell) improves* X, or X is *(Blackwell) dominated* by $X^\dagger$. When neither $X^\dagger \succ X$ nor $X \succ X^\dagger$, I say $X^\dagger$ and X are *not (Blackwell) ranked.*

After observing the joint realisation of (X, Z) for her applicant i, evaluator j choses whether to *approve* or *reject* him; $a_i^j \in \{Y, N\}$ respectively. Approvals have a type independent cost $c \in (0,1)$ and a payoff 1 if and only if the applicant has type $\theta = H$. Rejections are costless, and never yield payoffs. Formally, evaluator j's payoff from her decisions in period t will be given by:

$$\int_{i \in R_t^j} \mathbb{1}\{a_i^j = Y\}\left(\mathbb{1}\{\theta_i = H\} - c\right) di \tag{2.5}$$

where

$$R_t^j := \{\iota : \text{applicant } \iota \text{ applied to } j \text{ in period } t\}$$

and i are the labels evalautor j's applicants draw uniformly from the set R_t^j [7] .

Evaluator j's strategy is a mapping $\sigma_j : \{x_f, x_s\} \times \mathcal{Z} \to [0,1]$, assigning to every joint realization (x, z) of X and Z a probability that $a = Y$. A strategy profile $\{\sigma_j\}_{j=1}^K$ defines a

[6] Recall that $X^\dagger \succsim X$ and $X \succsim X^\dagger$ implies $X = X^\dagger$ so strict dominance is without loss of generality. See Escudé et al., 2022.

[7] Hence, observing the applicant's label i is not informative about her age, beyond the fact that $i \in R_t^j$.

steady state in the K-evaluation game. In a steady state, the mass of evaluator j's applicants with types θ and age τ is the same in every period, and is denoted by $m^j_{\theta,\tau}$.

This steady state admits a simple characterisation when the strategy profile is symmetric. My main discussion relies only on this symmetric characterisation. Nonetheless, for completeness, I describe the steady state under an arbitrary strategy profile $\{\sigma_j\}_{j=1}^K$ in the next paragraph.

For this complete characterisation, consider a permutation $\mathcal{P}_\iota \in S_K$ of the set $\{1, 2, ..., K\}$, where S_K is the symmetric group of order K[8]. Suppose applicant ι applies to evaluator $\mathcal{P}_\iota(\tau+1)$ when he is of age τ, unless he is approved by then. Such an applicant with type θ will still be in the applicant pool by age T with probability:

$$\mathcal{S}(\theta, T; \mathcal{P}_\iota) = \begin{cases} 1 & T = 0 \\ \prod_{\tau=0}^{T-1} \mathbb{E}_{X,Z}\left[1 - \sigma_{\mathcal{P}_\iota(\tau+1)}(x, z) \mid \theta, \tau\right] & T > 0 \end{cases}$$

Then, the mass of applicants with type θ and age τ who apply to evaluator j is given plainly by:

$$m^j_{L,\tau} := (1 - p_0) \times \mathbb{E}_{\mathcal{P}_\iota \in S_K}\left[\mathcal{S}(\theta, \tau; \mathcal{P}_\iota) \mid \mathcal{P}_\iota(\tau + 1) = j\right] \tag{2.6}$$

$$m^j_{H,\tau} := p_0 \times \mathbb{E}_{\mathcal{P}_\iota \in S_K}\left[\mathcal{S}(\theta, \tau; \mathcal{P}_\iota) \mid \mathcal{P}_\iota(\tau + 1) = j\right] \tag{2.7}$$

Evaluator j's expected payoff in a given period of the steady state is then given by:

$$(1 - c) \times \sum_{t=0}^{K-1} m^j_{H,t}\, \mathbb{E}_{X,Z}\left[\sigma_j(x, z) \mid \theta = H, \tau = t\right] - c \times \sum_{t=0}^{K-1} m^j_{L,t}\, \mathbb{E}_{X,Z}\left[\sigma_j(x, z) \mid \theta = L, \tau = t\right]$$

When the strategy profile $\{\sigma_j\}_{j=1}^K$ is *symmetric*; i.e. $\sigma_j = \sigma_k = \sigma$ for all evaluators j and k, all evaluators face an identical composition of applicants:

$$m^j_{L,\tau} = m_{L,\tau} := \begin{cases} K^{-1}(1 - p_0) & \tau = 0 \\ m_{L,\tau-1} \times \mathbb{E}_{X,Z}[1 - \sigma(x, z) \mid \theta = L, \tau] & \tau > 0 \end{cases} \tag{2.8}$$

$$m^j_{H,\tau} = m_{H,\tau} := \begin{cases} K^{-1}p_0 & \tau = 0 \\ m_{H,\tau-1} \times \mathbb{E}_{X,Z}[1 - \sigma(x, z) \mid \theta = H, \tau] & \tau > 0 \end{cases} \tag{2.9}$$

I refer to this as a symmetric steady state. A symmetric strategy profile $\{\sigma^*_j\}_{j=1}^K$ where σ^* is the strategy employed by all evaluators constitutes a *symmetric steady state equilibrium* (henceforth S-EQ) if:

$$\sigma^*(x, z) = \begin{cases} 0 & \mathbb{P}\left(\theta = H \mid m^*_\theta, X = x, Z = z\right) - c < 0 \\ \in [0, 1] & \mathbb{P}\left(\theta = H \mid m^*_\theta, X = x, Z = z\right) - c = 0 \\ 1 & \mathbb{P}\left(\theta = H \mid m^*_\theta, X = x, Z = z\right) - c > 0 \end{cases} \tag{2.10}$$

where $m^*_\theta = \left(m^*_{\theta,0}, m^*_{\theta,1}, ..., m^*_{\theta,K-1}\right)$ and $m^*_{\theta,\tau}$ is given by equations 2.8 and 2.9 when $\sigma = \sigma^*$. I focus on the S-EQ of the K-evaluation game.

[8]Recall that a permutation of the set $\{1, 2, ..., K\}$ is a bijection $\mathcal{P}$ from this set to itself. S_K is the symmetric group of order K; i.e. the set of all such permutations.

3 The Private K-Evaluation Game

In this section, I focus on the K-evaluation game with the uninformative history signal

$$Z_\emptyset : \{0, 1, ..., K-1\} \to \{\emptyset\}$$

$Z_\emptyset$ has the trivial distribution:

$$\mathbb{P}(Z_\emptyset(\tau) = \emptyset) = 1 \qquad \forall\, \tau \in \{0, 1, ..., K-1\}$$

I call this game the *private* K-evaluation game. Most of my paper focuses on this game as a benchmark. For brevity, I represent strategies with the following notation:

$$\sigma_f := \sigma(x_f, \emptyset) \qquad\qquad \sigma_s := \sigma(x_s, \emptyset)$$

$$\sigma = (\sigma_f, \sigma_s)$$

for a given type signal (x_f, x_s).

I start by showing that the private K-evaluation game is isomorphic to a game I call the *simultaneous* K-evaluation game. This isomorphism allows us to relate results about the private K-evaluation game to economic landscapes which, on first sight, fall outside our model. It also proves technically very useful, as it underlines most proofs in the paper.

In the *simultaneous* K-evaluation game, a candidate with unknown type $\theta \in \{L, H\}$ applies to K evaluators *simultaneously* for approval. Following his application, every evaluator costlessly and privately observes an IID realisation of the type signal X, where:

$$X \mid \theta \overset{\text{IID}}{\sim} F^\theta$$

as before. Following their respective observations, evaluators independently decide whether to offer approval or not to the applicant; $a \in \{Y, N\}$ respectively. Evaluator j's strategy $\sigma_j : \{x_f, x_s\} \to [0, 1]$ maps her signal realisation to the probability she will offer funding to the applicant. The candidate might receive none, one, or multiple approval offers. If he receives multiple, he chooses a single one to accept. He chooses whose offer to accept uniformly at random. An evaluator receives payoff:

$$u(\theta) = \mathbb{1}\{\theta = H\} - c$$

if her funding offer is accepted by the candidate of type θ.

Proposition 3.1. *Given the same type signal X and primitives p_0 and c, the simultaneous K-evaluation game is isomorphic to the private K-evaluation game. i.e, given the same evaluator strategies $\{\sigma_j\}_{j=1}^K$:*

1. *An applicant with type θ is funded by evaluator j with the same probability.*
2. *Evaluator j's expected payoff in a period of the steady state of the private K-evaluation game equals his expected payoff in the simultaneous K-evaluation game.*

Proof. See Section 6. □

The proof draws an equivalence between applicant ι's application order $\mathcal{P}_\iota \in S_K$ in the private K-evaluation game, and his tie-breaking order between the approvals he receives in the simultaneous K-evaluation game.

Proposition 3.1 helps interpret the private K-evaluation game in a broader economic context. In many admission processes, candidates apply to multiple evaluators *simultaneously*. For instance, prospective students apply to different graduate study programs all at once. Likewise, graduating students apply to different employers in the same industry during the same recruitment cycle, likely before any employer announces her decision. Such admission processes do not admit immediate interpretation through the K-evaluation game. However, Proposition 3.1 establishes that to the extent they resemble the *simultaneous* K-evaluation game, they can be viewed through the lens of a private K-evaluation game. Beyond expanding the economic relevance of our model, Proposition 3.1 also proves technically very useful. By converting a dynamic model to essentially a static one, it facilitates our analysis greatly.

3.1 S-EQ in the Private K-Evaluation Game

In the steady state of any K-evaluation game, an evaluator's applicant pool consists of newly born candidates as well as those with previous rejections. Therefore, the probability with which an evaluator faces an applicant of type $\theta = H$ in the steady state differs from the *fresh prior* p_0. I call the probability that a randomly picked candidate in the applicant pool has type $\theta = H$ the *pool prior*:

Definition 3.2. For an arbitrary strategy profile $\{\sigma_j\}_{j=1}^K$, the *pool prior* δ is defined as:

$$\delta := \frac{\sum_{j=1}^{K} \sum_{\tau=0}^{K-1} m_{H,\tau}^j}{\sum_{j=1}^{K} \sum_{\tau=0}^{K-1} m_{H,\tau}^j + m_{L,\tau}^j} \tag{3.1}$$

where vectors $m_{\theta,\tau}^j$ are given by equations 2.6 and 2.7.

Definition 3.3. The pool prior under the S-EQ of the K-evaluation game is called the *equilibrium pool prior*, and is denoted as δ^*:

$$\delta^* = \frac{\sum_{\tau=0}^{K-1} m_{H,\tau}^*}{\sum_{\tau=0}^{K-1} m_{L,\tau}^*} \tag{3.2}$$

where $m_{\theta,\tau}^*$ are given by equations 2.8 and 2.9 when σ^* is evaluators' equilibrium strategies.

Note that in the private K-evaluation game with a symmetric strategy profile, the expressions for $m_{\theta,\tau}$ take an especially simple form:

$$m_{L,\tau} := K^{-1}(1-p_0)\left[\mu_0(1-\sigma_f) + (1-\mu_0)(1-\sigma_s)\right]^\tau \tag{3.3}$$

$$m_{H,\tau} := K^{-1}p_0\left[(1-\mu_1)(1-\sigma_f) + \mu_1(1-\sigma_s)\right]^\tau \tag{3.4}$$

When evaluators' strategies are symmetric in the K-evaluation game, they face identical compositions of applicants. Hence, the pool prior δ in a symmetric steady state also describes the probability that any given evaluator faces an applicant of type $\theta = H$.

The *pool prior* is endogenously determined, unlike the *fresh prior*. In an S-EQ, X and σ^* determine δ^* through m_θ^*. Simultaneously, σ^* must be optimal given δ^*:

$$\sigma_f^* \begin{cases} = 0 & \frac{\delta^*}{1-\delta^*} \times \frac{x_f}{1-x_f} < \frac{c}{1-c} \\ \in [0,1] & \frac{\delta^*}{1-\delta^*} \times \frac{x_f}{1-x_f} = \frac{c}{1-c} \\ = 1 & \frac{\delta^*}{1-\delta^*} \times \frac{x_f}{1-x_f} > \frac{c}{1-c} \end{cases} \qquad \sigma_s^* \begin{cases} = 0 & \frac{\delta^*}{1-\delta^*} \times \frac{x_s}{1-x_s} < \frac{c}{1-c} \\ \in [0,1] & \frac{\delta^*}{1-\delta^*} \times \frac{x_s}{1-x_s} = \frac{c}{1-c} \\ = 1 & \frac{\delta^*}{1-\delta^*} \times \frac{x_s}{1-x_s} > \frac{c}{1-c} \end{cases} \tag{3.5}$$

Alongside 2.1, 3.5 implies that whenever $a = Y$ is optimal following a *Fail* signal, it is also optimal following a *Success* signal. Simply put, a *Fail* signal is evidence for the *Low* type, and a *Success* signal for the *High* type; hence an evaluator is more willing to approve following the latter. Hence, we necessarily have $\sigma_s^* \geq \sigma_f^*$ in equilibrium; *Fail* signals will lead to more frequent rejections than *Success*es.

Approved candidates retire immediately, while rejected candidates remain in the applicant pool granted they have more evaluators to apply to. As *Succeed*ing candidates retire faster than *Fail*ing candidates in any equilibrium, an *adverse selection* problem emerges in the applicant pool. This drives the pool prior below the fresh prior in any equilibrium; $\delta^* \leq p_0$. Coupled with condition O-F on the type signal X, we then get:

$$\frac{\delta^*}{1-\delta^*} \times \frac{x_f}{1-x_f} \leq \frac{p_0}{1-p_0} \times \frac{x_f}{1-x_f} \leq \frac{c}{1-c} \tag{3.6}$$

meaning $\sigma_f^* = 0$ always constitutes an equilibrium in the private K-evaluation game. Lemma 3.4 follows this reasoning, and establishes the stronger fact that $\sigma_f^* = 0$ is *necessary* in *any* equilibrium of the private K-evaluation game.

Lemma 3.4. *In any steady state equilibrium of the private K-evaluation game, $\sigma_j^*(x_f) = 0$ for all evaluators $j \in \{1, 2, ..., K\}$.*

Proof. See Section 6. □

Evaluators "cherry pick": while they approve and take *Succeed*ing applicants, they let *Fail*ing applicants to remain in the pool and keep applying. Our intuition suggests that the adverse selection problem this generates would be ameliorated if evaluators approved *Succeed*ing applicants at a lower frequency. Lemma 3.5 confirms this intuition.

Lemma 3.5. *Let the type signal be the experiment (x_f, x_s) in the private K-evaluation game, and suppose $\sigma_j = \sigma = (0, \sigma_s)$ for all evaluators j. Where $\delta(\sigma_s) : [0,1] \to [0,1]$ is the mapping from σ_s to the induced pool prior:*

1. *$\delta(\sigma_s)$ is continuous and strictly decreasing.*
2. *$\delta(\sigma_s) \leq p_0$, with equality if and only if $\sigma_s = 0$.*

With the help of Lemmas 3.4 and 3.5, I prove that the S-EQ of the private K-evaluation game exists, and is unique.

Proposition 3.6. *Take the private K-evaluation game with primitives c and p_0, and type signal $(x_f, x_s) \in \Omega^{p_0,c}$. The* S-EQ *of this game exists and is unique. Evaluators' equilibrium strategies (σ_f^*, σ_s^*) satisfy:*

$$\sigma_f^* = 0 \qquad\qquad \sigma_s^* \in (0, 1]$$

Proof. As a consequence of Lemma 3.4, $\sigma_f^* = 0$ necessarily in any S-EQ. Recall from Lemma 3.5 that $\delta(\sigma_s)$ is a continuously decreasing function. Suppose we have:

$$\frac{\delta(1)}{1-\delta(1)}\frac{x_s}{1-x_s} - \frac{c}{1-c} \geq 0 \tag{3.7}$$

Then, due to 2.10, $\sigma_s^* = 1$ constitutes an S-EQ. Conversely, no $\sigma_s^* < 1$ can be part of an S-EQ then, since:

$$\frac{\delta(\sigma_s)}{1-\delta(\sigma_s)}\frac{x_s}{1-x_s} > \frac{\delta(1)}{1-\delta(1)}\frac{x_s}{1-x_s} \geq \frac{c}{1-c} \quad \forall\ \sigma_s < 1 \tag{3.8}$$

Now let:

$$\frac{\delta(1)}{1-\delta(1)}\frac{x_s}{1-x_s} - \frac{c}{1-c} < 0 \tag{3.9}$$

Then, $\sigma_s^* = 1$ cannot constitute an S-EQ. Observe that $\sigma_s = 0$ can never constitute an S-EQ, since Lemma 3.5 and Assumption 2.1 would then imply:

$$\frac{\delta(0)}{1-\delta(0)}\frac{x_s}{1-x_s} = \frac{p_0}{1-p_0}\frac{x_s}{1-x_s} > \frac{c}{1-c} \tag{3.10}$$

Hence, when 3.9 holds, we must have $\sigma_s^* \in (0, 1)$. From 3.5, we know that:

$$\frac{\delta^*}{1-\delta^*}\frac{x_s}{1-x_s} - \frac{c}{1-c} = 0 \tag{3.11}$$

must be satisfied in such an S-EQ. But due to Lemma 3.5, there is a unique value of $\sigma_s^* \in (0, 1)$ which ensures 3.11 holds. Therefore, the equilibrium $\sigma_s^* \in (0, 1)$ exists, and constitutes the unique S-EQ.

□

In any S-EQ, the adverse selection problem in the pool pulls the pool prior δ below p_0, imposing $\sigma_f^* = 0$. σ_s^* and δ^*, the equilibrium extent of adverse selection, are determined jointly. Recall that Lemma 3.5 defined $\delta(1)$ to be the pool prior when all evaluators employ $\sigma = (0, 1)$. $\sigma^* = (0, 1)$ and $\delta^* = \delta(1)$ if and only if funding every *Success* applicant yields non-negative payoff even when all other evaluators do the same:

$$\frac{\delta(1)}{1-\delta(1)}\frac{x_s}{1-x_s} \geq \frac{c}{1-c} \tag{3.12}$$

When 3.12 is violated, the severity of adverse selection in the pool precludes $\sigma_s^* = 1$. Restoring equilibrium requires *alleviating* it. This is achieved through decreasing the probability evaluators approve upon favourable evidence; σ_s^* decreases below 1. In this unique S-EQ, we then have:

$$\frac{\delta^*}{1-\delta^*}\frac{x_s}{1-x_s} = \frac{c}{1-c} \tag{3.13}$$

where $\delta = \delta^*(\sigma_s)$, and $\sigma_s^* \in (0,1)$ is the unique value ensuring 3.13 holds. In such an equilibrium, evaluators are *indifferent* between accepting and rejecting an applicant with a *Success* signal. They accept only a positive fraction of them, and expect zero payoff from doing so.

This insight reveals that $\sigma^* = (0,1)$ as long as evaluators' expected payoffs are non-negative when they all employ this strategy. Evaluators' equilibrium payoffs are zero otherwise. Hence, equilibrium payoffs are fully pinned down by inspecting what evaluators' payoffs would be if they all employed strategies $\sigma = (0,1)$.

Lemma 3.7. *For given primitives p_0 and c in the private K-evaluation game, let:*

$$\pi^*(x_f, x_s) : \Omega^{p_0,c} \to \mathbb{R}$$

be the mapping from an individually obedient type signal to evaluators' equilibrium payoffs. Likewise, let:

$$\pi(x_f, x_s) : [0, 0.5] \times [0.5, 1] \to \mathbb{R}$$

be the mapping from any experiment to evaluators' payoffs when $\sigma_j = (0,1) \quad \forall j \in \{1, 2, ..., K\}$. Then:

$$\pi^*(x_f, x_s) = \max\{0, \pi(x_f, x_s)\}$$

Proof. See Section 6. □

3.2 Improving Type Signals in the Private K-Evaluation Game

I now turn to our central question. How do evaluators' equilibrium payoffs change when the type signal X is replaced by $X^\dagger$ where $X^\dagger \succ X$? I begin with a straightforward corollary to Blackwell's Theorem, describing when $X^\dagger \succ X$ for two binary experiments $X^\dagger$ and X.

Corollary 3.8. Take the type signals X and $X^\dagger$, with signal realisations $(x_f, x_s) \neq (x_f^\dagger, x_s^\dagger)$ respectively. Then:

$$X^\dagger \succ X \quad \iff \quad x_f \geq x_f^\dagger,\ x_s^\dagger \geq x_s$$

Proof. See Section 6. □

Consider an evaluator who assigns a probability $\delta = \mathbb{P}(\theta = H)$ to her applicant's type being *High*. Where her type signal is (x_f, x_s), she updates her posterior probability to:

$$\mathbb{P}(\theta = H \mid x = x_f) = \frac{\delta x_f}{\delta x_f + (1-\delta)(1-x_f)} < \delta$$

upon observing the signal realisation x_f. This posterior is decreasing in the value of x_f. Upon observing x_s, she updates to:

$$\mathbb{P}(\theta = H \mid x = x_s) = \frac{\delta x_s}{\delta x_s + (1-\delta)(1-x_s)} > \delta$$

which is increasing in the value of x_s. A *Fail* (*Success*) realisation is evidence for the *Low* (*Success*) type for an evaluator. Hence, I call a decrease (increase) in x_f (x_s) a strengthening of bad (good) news. Corollary 3.8 establishes that a type signal $X^\dagger$ Blackwell dominates X either when (i) it delivers stronger bad news, and/or (ii) it delivers stronger good news.

Before I study how a more informative type signal affects S-EQ payoffs in the private K-evaluation game, I demonstrate what the answer would be if we had a *single* evaluator. Through Blackwell's Theorem, we already know a more informative type signal must be weakly beneficial in this single person decision problem. Lemma 3.9 takes this a step further: when there is a single evaluator, she *strictly* benefits from a more informative type signal, provided the improved signal is individually obedient.

Lemma 3.9. *Suppose $K = 1$ in the private K-evaluation game. Let $X \succ X'$ and $X \in \Omega^{p_0,c}$. The steady state expected payoff of an evaluator in a period is higher under X than under X'.*

Proof. See Section 6. □

When the number of evaluators exceeds one, evaluators face and approve applicants of different ages in a given period of the steady state. This can complicate reasoning about evaluators' steady state payoffs. To simplify this problem, I leverage the equivalence between the private and simultaneous K-evaluation games, established in Proposition 3.1. Through this equivalence, I convert this dynamic problem to an essentially static one. Lemma 3.10 uses this equivalence, and derives $\pi(x_f, x_s)$; what evaluators' payoffs would be if they all employed $\sigma = (0, 1)$.

Lemma 3.10. *Let (x_f, x_s) be the type signal in the private K-evaluation game. Then:*

$$\begin{aligned}\pi(x_f, x_s) = K^{-1} \Big[&(p_0 - c) + (1 - p_0) \times c \times \mu_0(x_f, x_s)^K \\ &- p_0 \times (1 - c) \times (1 - \mu_1(x_f, x_s))^K \Big]\end{aligned} \tag{3.14}$$

Proof. See Section 6. □

Lemma 3.10 relies on a simple insight. In a symmetric steady state, evaluators expect equal payoffs every period. Thus, inspecting the *sum* of their payoffs suffices. This is pinned down by the candidates who receive admission from at least one evaluator in a given period. When all evaluators employ the strategy $\sigma = (0, 1)$, all candidates except those who draw K consecutive *Fail* signals are admitted by at least one evaluator. Thus, the sum of evaluators' payoffs are given simply by their net gains of admitting these candidates. This insight simplifies the problem greatly. However, the complication that one needs to track candidates receiving admission multiple periods after their births remains. It is this complication which is resolved by Proposition 3.1.

With Lemma 3.7 in mind, I first establish how Blackwell improvements of X affect $\pi(x_f, x_s)$ to uncover the influence of Blackwell improvements on evaluators' equilibrium payoffs $\pi^*(x_f, x_s)$.

Lemma 3.11. *Given the primitives p_0 and c of the private K-evaluation game, $\pi(x_f, x_s)$:*

1. *strictly increases in x_s.*
2. *is strictly quasi-concave in x_f.*
3. *given x_s, is maximised uniquely at the value $x_f^*(x_s) \in (0, 0.5)$ solving:*

$$\frac{c}{1-c} = \frac{p_0}{1-p_0} \frac{x_s}{1-x_s} \left(\frac{x_f^*(x_s)}{1 - x_f^*(x_s)} \right)^{K-1} \tag{3.15}$$

Proof. See Section 6. □

Lemma 3.11 reveals an asymmetry. Holding the strength of bad news (x_f) constant, stronger good news always increases $\pi(x_f, x_s)$. However, holding the strength of good news constant, stronger bad news do not have a monotone effect on $\pi(x_f, x_s)$. While $\pi(x_f, x_s)$ initially increases, it starts decreasing once x_f falls below a threshold, $x_f^*(x_s)$. This threshold is always in the interior, $x_f^*(x_s) \in (0, 0.5)$, and depends on the strength of good news (x_s).

For a given strength of good news, evaluators' equilibrium payoffs are maximised when the strength of bad news matches this threshold; $x_f = x_f^*(x_s)$. This optimal type signal $X^* = (x_f^*(x_s), x_s)$ admits a natural interpretation. To convey it, I introduce a notion of *relevance of history* in the K-evaluation game.

Suppose the type signal (x_f, x_s) is such that an evaluator observing a *Success* signal would approve her applicant *even if* she knew $K-1$ evaluators saw *Fail* signals previously. Then, I say *history is irrelevant* under type signal X in the K-evaluation game with primitives p_0 and c.

Definition 3.12. *History is irrelevant* under experiment (x_f, x_s) in the K-evaluation game with primitives p_0 and c if:

$$\frac{p_0}{1-p_0}\left(\frac{x_f}{1-x_f}\right)^{K-1}\frac{x_s}{1-x_s} \geq \frac{c}{1-c}$$

When X^* is improved through stronger bad news, history gains relevance. Thus, among signals with a fixed strength of good news, X^* is the *most informative* type signal for which history is irrelevant. Lemma 3.11 establishes that provided history is relevant, *any* strengthening of bad news decreases $\pi(x_f, x_s)$, unless accompanied by a corresponding strengthening of good news.

Bridging Lemmas 3.7 and 3.11 brings us to Theorem 1. Theorem 1 characterises how equilibrium payoffs in the private K-evaluation game respond to Blackwell improvements in the type signal X.

Theorem 1. *Given the primitives p_0 and c, evaluators' equilibrium payoff in the private K-evaluation game, $\pi^*(x_f, x_s)$ is:*

1. *weakly increasing in x_s for a given x_f.*
2. *quasi-concave in x_f for a given x_s.*

furthermore, for a given x_s:

3. *is maximised uniquely at $x_f = \min\{x_f^*(x_s), B_f\}$.*
4. *is strictly increasing in x_f, for $x_f \in [\chi_f, \min\{x_f^*(x_s), B_f\}]$, for some $\chi_f \in [0, \min\{x_f^*(x_s), B_f\})$.*

Below, I provide a proof sketch for the Theorem. Section 6 fills the gaps in the sketched proof.

Proof Sketch: The first two claims follow straightforwardly from Lemma 3.11 and:

$$\pi^*(x_f, x_s) = \max\{0, \pi(x_f, x_s)\}$$

established in Lemma 3.7.

For the third claim, note that we are concerned with the solution to the problem:

$$\max_{0 \leq x_f \leq B_f} \pi^*(x_f, x_s) \tag{3.16}$$

where $\pi^*(x_f, x_s) = \max\{0, \pi(x_f, x_s)\}$. I first inspect the solution to the problem:

$$\max_{0 \leq x_f \leq B_f} \pi(x_f, x_s) \tag{3.17}$$

Through Lemma 3.7, the solution to 3.17 is clearly $x_f^*(x_s)$ when $x_f^*(x_s) \leq B_f$. Otherwise, B_f is in the region where $\pi(x_f, x_s)$ is an increasing function of x_f; hence the maximiser is given by B_f. Establishing that as a function of x_f, $\pi(x_f, x_s)$ is maximised at a strictly positive value:

$$\pi^* \left(\min\{x_f^*(x_s), B_f\}, x_s\right) = \pi \left(\min\{x_f^*(x_s), B_f\}, x_s\right) > 0 \tag{3.18}$$

concludes the proof. The fourth claim follows simply from inequality 3.18, and the observation that $\pi(x_f, x_s)$ is continuous in both arguments.

□

The influence of the type signal on the gravity of adverse selection, as might be guessed, is the main driver of this non-monotonicity. In a standard single person decision problem, more information unambiguously improves the decision-maker's expected payoff. It does so by helping her tailor her action further to the unknown state. This *direct* informational benefit is still present in the K-evaluation game. Taking the composition of the applicant pool as given, evaluators would expect a higher payoff under a more informative type signal X. If we had a single evaluator, this would be the only effect present, and as Lemma 3.9 makes clear, any improvement in the type signal would increase our evaluator's payoff.

However, a change in the type signal X also has an *indirect* effect on evaluators' payoffs through its influence on the adverse selection in the applicant pool. By determining evaluators' approval decisions, X strongly influences valuators' applicant pool, hence δ. Hence, improving evaluators' information affects both their ability to respond to uncertainty, and *the uncertainty they face itself.* The more δ^* falls below p_0, the more equilibrium adverse selection is amplified, and evaluators' equilibrium payoffs are lowered.

The net effect of a change in X on evaluators' equilibrium payoffs depends both on its *direct* informational effect, and its *indirect* effect on the composition of the applicant pool. Lemma 3.13 shows that for constant x_s, the pool prior δ is monotone decreasing in x_f in *any* private K-evaluation game.

Lemma 3.13. *Let δ^* and $\tilde{\delta}^*$ be the equilibrium pool priors in the private K-evaluation game under type signals (x_f, x_s) and $(\tilde{x}_f, x_s)$. Then:*

$$\tilde{x}_f \leq x_f \qquad \Longrightarrow \qquad \tilde{\delta}^* \leq \delta^* \tag{3.19}$$

Inequality 3.19 holds strictly if and only if $\pi^*(x_f, x_s) = 0$.

Lemma 3.13 establishes that strengthening bad news always has a negative indirect effect. Theorem 1 shows that this negative indirect effect always dominates the positive direct effect, unless history is irrelevant. Although evaluators receive better information about their applicants, these informational benefits are offset as they expect a worse applicant pool.

Which changes in the type signal X guarantee a weak improvement in evaluators' equilibrium payoffs, then? To answer this question, I construct an auxiliary decision problem with a single decision maker. This auxiliary problem also provides additional insight into Theorem 1.

For the private K-evaluation game with fresh prior p_0, funding cost c, and type signal X, our *auxiliary* decision problem proceeds as follows: an *auxiliary evaluator* faces a single applicant with private type $\theta \in \{L, H\}$. She chooses whether to approve or reject him, $a \in \{Y, N\}$ respectively. Her payoff is given by:

$$u^A(\theta, a) = \mathbb{1}\{a = Y\}(\mathbb{1}\{\theta = H\} - c)$$

θ has the prior distribution $\mathbb{P}(\theta = H) = p_0$. First, the applicant draws K realisations of X, which are IID conditional on her type θ. These K draws are not observable to the auxiliary evaluator. Instead, she observes their *K-garbling* χ before her decision:

Definition 3.14. Let $\{x_1, ..., x_K\}$ be K IID draws of the type signal X, from an applicant with type θ. The random variable χ with realisations $\xi \in \{\xi_f, \xi_s\}$ is called the *K-garbling of X*, if:

$$\xi = \begin{cases} \xi_f & \text{if } x_1 = ... = x_K = x_f \\ \xi_s & \text{if } \exists\ i \in \{1, 2, ..., K\} \text{ s.t. } x_i \neq x_f \end{cases}$$

The signal realisations ξ_f and ξ_s are labeled after the *likelihood ratios* attached to them:

$$\frac{\mathbb{P}(\xi = \xi_f \mid \theta = H)}{1 - \mathbb{P}(\xi = \xi_f \mid \theta = H)} = \frac{\xi_f}{1 - \xi_f} \qquad \frac{\mathbb{P}(\xi = \xi_s \mid \theta = H)}{1 - \mathbb{P}(\xi = \xi_s \mid \theta = H)} = \frac{\xi_s}{1 - \xi_s}$$

χ garbles the information provided by K IID draws of X. It only reveals whether *all* K draws were *Fail*s, or there was at least one *Success*.

Note that the conditional probabilities describing χ are:

$$\mathbb{P}(\xi = \xi_f \mid \theta = L) = \mu_0^K \qquad \mathbb{P}(\xi = \xi_f \mid \theta = H) = (1 - \mu_1)^K \tag{3.20}$$

Following the convention we adapted for the type signal, the signal realisations ξ_f and ξ_s are labeled after the respective likelihood ratios attached to them. Through 3.20, we have:

$$\frac{\xi_f}{1 - \xi_f} = \left(\frac{1 - \mu_1}{\mu_0}\right)^K \qquad \frac{\xi_s}{1 - \xi_s} = \frac{1 - (1 - \mu_1)^K}{1 - \mu_0^K}$$

The K-garbling χ of the type signal X is certainly less informative than observing all K signals the candidate drew. It also delivers stronger bad news than a single draw of the type signal X; since it reveals that a candidate drew K *Fail* signals. Nonetheless, χ is not necessarily *more* informative than X. It only reveals whether a candidate got *at least* one *Success* signal. This can be less encouraging than a single *Success* signal from X, as it doesn't rule out the possibility that the candidate's *Success* signal was followed by multiple *Fail* signals.

As an example, consider the type signal with:

$$x_f = 1/3 \qquad x_s = 2/3$$

hence:

$$\mu_0 = \mu_1 = 2/3$$

The associated 2-garbling delivers stronger bad news, as it reveals *two Fail* signals:

$$\frac{\xi_f}{1-\xi_f} = \left(\frac{1}{2}\right)^2 \qquad \Longrightarrow \qquad \xi_f = \frac{1}{5}$$

However, it delivers weaker good news; since ξ_s is indicative of two *Success* signals, or one success and one *Fail* signal:

$$\frac{\xi_s}{1-\xi_s} = \frac{1-(1/3)^2}{1-(2/3)^2} \qquad \Longrightarrow \qquad \xi_s = \frac{8}{13}$$

How do improvements in the type signal X and its K-garbling χ relate? Improving X through stronger bad news has a twofold effect on χ. On the one hand, it strengthens bad news from χ, as it makes K *Fail* draws stronger evidence for a *Low* type. On the other hand, it also affects the strength of good news from χ. The news that a candidate received at least one *Success* signal might not be much relief for an evaluator when the *Fail* signal is strongly damning. This intuition suggests that improving X through stronger bad news might not improve its K-garbling; as it weakens the good news it delivers.

Proposition 3.15 confirms this intuition. It also delivers a positive conclusion: improving X through stronger good news *does* improve χ. While it does not affect the strength of bad news χ delivers, it makes good news from it more encouraging.

Proposition 3.15. *Let χ and $\chi^\dagger$ be the K-garblings of the type signals (x_f, x_s) and $(x_f^\dagger, x_s^\dagger)$ respectively. Then:*

$$x_f^\dagger = x_f \text{ and } x_s^\dagger > x_s \qquad \Longrightarrow \qquad \chi^\dagger \succ \chi$$

On the other hand, $\chi^\dagger$ and χ cannot be Blackwell ranked when $x_f^\dagger < x_f$ and $x_s^\dagger = x_s$.

Proposition 3.15 elucidates that improving the type signal X does not suffice for an improvement in its K-garbling, χ. Completing our inquiry, Proposition 3.16 establishes that this condition is nonetheless necessary. χ can deliver stronger bad news only if X does. However, stronger bad news from X decrease the strength of good news from χ. To counteract this, X must also deliver stronger good news.

Proposition 3.16. *Let X and $X^\dagger$ be two distinct binary Blackwell experiments, and χ and $\chi^\dagger$ be their K-garblings respectively. Then:*

$$X^\dagger \succ X \not\Longrightarrow \chi^\dagger \succ \chi$$

However:

$$\chi^\dagger \succ \chi \Longrightarrow X^\dagger \succ X$$

Proof, Proposition 3.16: The first claim is a corollary to Proposition 3.15. For the second claim, note that $\chi^\dagger \succ \chi$ implies:

$$\xi_f^\dagger \leq \xi_f \qquad\qquad \xi_s^\dagger \geq \xi_s$$

Since:

$$\frac{\xi_f}{1-\xi_f} = \left(\frac{x_f}{1-x_f}\right)^K \qquad\qquad \frac{\xi_f^\dagger}{1-\xi_f^\dagger} = \left(\frac{x_f^\dagger}{1-x_f^\dagger}\right)^K$$

it must be that $x_f^\dagger \leq x_f$. But the proof for Proposition 3.15 shows that this implies $\xi_s^\dagger \leq \xi_s$ unless $x_s^\dagger > x_s$.

□

The auxiliary decision problem relates closely to the private K-evaluation game. As Proposition 3.17 establishes, the sum of evaluators' equilibrium payoffs given their type signal X equals the auxiliary evaluator's expected payoff under the K-garbling χ.

Proposition 3.17. *Let $\pi^*(x_f, x_s)$ be an evaluator's equilibrium payoff in the original problem. Then:*

$$\pi^A(\xi_f, \xi_s) = K\pi^*(x_f, x_s)$$

In the auxiliary problem, rejections can be based only on whether the applicant produced K *consecutive Fail* signals in the binary experiment X or not. As X satisfies the obedience constraints, it is always optimal to reject an applicant with such signals. Funding the applicant otherwise is optimal if and only if the expected payoff from doing so given she had at most $K-1$ *Fail* signals is non-negative. When this condition fails, the auxiliary evaluator rejects every applicant, collecting zero payoff.

Whether funding an applicant with at most $K-1$ *Fail* signals yields positive expected payoff also determines the unique equilibrium of the private K-evaluation game. $\sigma^* = (0, 1)$ if and only if this payoff is non-negative. When such, all applicants except the ones with K consecutive *Fail* signals are funded, and this determines the total payoff of the evaluators. Otherwise, evaluators mix between funding and rejecting applicants with *Success* signals. However, this yields the same payoff to them as funding no applicant would, since their indifference implies they expect zero payoff in equilibrium.

This connection helps us study the effect of changing evaluators' type signals through a single person decision problem. Proposition 3.17 implies that $\pi^*(x_f, x_s)$ increases following a change in X if and only if the corresponding change in the K-garbling χ increases $\pi^A(\xi_f, \xi_s)$. But the auxiliary evaluator simply faces a single person decision making problem with two states and two actions. Therefore, $\chi^\dagger$ is more valuable than χ in the auxiliary problem with arbitrary p_0 and c if and only if $\chi^\dagger \succ \chi$[9]. This leads us to Corollary 3.18.

Corollary 3.18. Consider the two binary Blackwell experiments (x_f, x_s) and $(x_f^\dagger, x_s^\dagger)$, and their respective K-garblings χ and $\chi^\dagger$. Then:

$$\pi^*(x_f^\dagger, x_s^\dagger) \geq \pi^*(x_f, x_s) \quad \iff \quad \chi^\dagger \succ \chi$$

3.3 Optimal Strategies in the Private K-Evaluation Game

Theorem 1 establishes that improvements in evaluators' type signal might decrease their equilibrium payoffs in the private K-evaluation game. This owes to a *rejection externality*: an evaluator's rejection decisions impacts others' applicant pools. The type signal X influences the

[9] Blackwell's Theorem states that $\chi^\dagger$ is more valuable than χ in *any* decision problem if and only if $\chi^\dagger \succ \chi$. However, Blackwell and Girshick, 1954, in the proof of Theorem 12.4.1, show that another equivalent condition is for $\chi^\dagger$ to be more valuable than χ in any single person *two-state two-action* decision problem.

severity of the adverse selection this externality concocts. In this Section, I study how a social planner who could control evaluators' strategies could mitigate this externality.

I retain focus on the private K-evaluation game. A social planner chooses evaluators' strategy profile, given their type signal X. His objective is to maximise the sum of all evaluators' payoffs. I call the strategy profile the planner chooses the *optimal* strategy profile.

Throughout, we focused on a symmetric equilibrium. In this Section, I first study the planner's problem when he is constrained to *symmetric* strategy profiles. I compare this *symmetric optimal* strategy profile to the equilibrium strategy profile, and describe how it increases evaluators' payoffs relative to the S-EQ.

Then, I describe the optimal *asymmetric* strategy profile, where the planner assigns a different strategy to every evaluator. The optimal *asymmetric* strategy profile is remarkably simple and has two features. First, only a subset of evaluators approve any applicants; I call them the *active* evaluators. Second, the active evaluators approve all and only applicants who *Succeed*. Hence, it replicates a private k^*-evaluation game, where k^* is the number of evaluators left active.

Formally, for $i \in \{1, 2, ..., K\}$, let the function:

$$\Pi_i(x_f, x_s; \sigma_i, \sigma_{-i}) : \Omega^{p_0,c} \times [0,1]^{2K} \to \mathbb{R} \tag{3.21}$$

be evaluator i's steady state payoff when evaluators' type signal is $(x_f, x_s) \in \Omega^{p_0,c}$ and evaluator j follows strategy $\sigma_j = (\sigma_j(f), \sigma_j(s))$. First, I study the optimal symmetric strategy profile:

$$\sigma^{SP} \in \arg\max_{\sigma \in [0,1]^2} \sum_{i=1}^{K} \Pi_i(x_f, x_s; \sigma_i, \sigma_{-i}) \quad \text{s.t. } \sigma_i = \sigma \quad \text{for all } i \in \{1, 2, ..., K\} \tag{SP}$$

I denote the value of this problem as V^{SP}:

$$V^{SP}(x_f, x_s) := \max_{\sigma \in [0,1]^2} \left\{ \sum_{i=1}^{K} \Pi_i(x_f, x_s; \sigma_i, \sigma_{-i}) \quad | \, \sigma_i = \sigma \quad \forall \, i \in \{1, 2, ..., K\} \right\}$$

Theorem 2 describes the optimal symmetric strategy profile, and evaluators' payoffs under it. When history is irrelevant, the optimal symmetric strategy profile mirrors evaluators' equilibrium strategies, and their payoffs equal their equilibrium payoffs. When history is relevant, the optimal symmetric strategy profile instructs evaluators to reject some applicants upon a *Success* signal. Evaluators' payoffs equal their equilibrium payoffs under the type signal $(x_f^*(x_s), x_s)$.

Theorem 2. *Given the primitives p_0, and c:*

$$V^{SP}(x_f, x_s) = \begin{cases} K \times \pi(x_f, x_s) & x_f \geq x_f^*(x_s) \\ K \times \pi(x_f^*(x_s), x_s) & x_f < x_f^*(x_s) \end{cases}$$

V^{SP} is achieved by a unique symmetric strategy profile $\sigma^{SP} = (\sigma_f^{SP}, \sigma_s^{SP})$.

$$\sigma_f^{SP} = 0 \quad \forall \, (x_f, x_s) \in \Omega^{p_0,c} \qquad\qquad \sigma_s^{SP} = \begin{cases} 1 & x_f \geq x_f^*(x_s) \\ < 1 & x_f < x_f^*(x_s) \end{cases}$$

Proof. Take the experiment (x_f, x_s) and symmetric strategy profile $\sigma_i = \sigma = (\sigma_f, \sigma_s) \forall\ i \in \{1, 2, ..., K\}$. Let $\tilde{\mu_0}$ ($\tilde{\mu_1}$) be the probability that an applicant of type $\theta = L$ ($\theta = H$) is rejected (funded) by the evaluator she applies to:

$$\tilde{\mu_0} = (1 - \sigma_f)\mu_0 + (1 - \sigma_s)(1 - \mu_0) \qquad \tilde{\mu_1} = \sigma_f(1 - \mu_1) + \sigma_s\mu_1 \tag{3.22}$$

The reasoning underlining Proposition 3.10 implies that:

$$\Pi_i(x_f, x_s, \sigma_i, \sigma_{-i}) = K^{-1}\left[p_0 - c + (1 - p_0)\tilde{\mu_0}^K - p_0(1 - \tilde{\mu_1})^K\right] \tag{3.23}$$

Corollary 5.5 to Blackwell's Theorem asserts that there exists a binary Blackwell experiment X' such that $X \succ X'$ and:

$$\mathbb{P}(x' = x'_f \mid \theta = L) = \tilde{\mu_0} \qquad \mathbb{P}(x' = x'_s \mid \theta = H) = \tilde{\mu_1}$$

Under the symmetric strategy profile $\sigma' = (0, 1)$, X' would provide the same payoff $\Pi_i(x_f, x_s, \sigma_i, \sigma_{-i})$ in 3.23. Therefore, the value of the objective:

$$\max_{X':\ X \succ X'} \sum_{j=1}^{K} \Pi_j(x_f, x_s, \sigma_i, \sigma_{-i}) \quad \text{such that } \sigma_i = (0, 1)\ \forall\ i \in \{1, 2, ..., K\}$$

is an upper bound for V^{SP}. From Lemma 3.11, and symmetry, this maximum is achieved when $\Pi_i(x_f, x_s, \sigma_i, \sigma_{-i}) = \pi(x_f, x_s)$ for all i if $x_f \geq x_f^*(x_s)$, and $\Pi_i(x_f, x_s, \sigma_i, \sigma_{-i}) = \pi(x_f^*(x_s), x_s)$ if $x_f < x_f^*(x_s)$.

Under the type signal X' and strategy $\sigma' = (0, 1)$, an evaluator rejects a *Low* type with probability $\tilde{\mu}_0$ and approves a *High* type with probability $\tilde{\mu}_1$. The same Corollary to Blackwell's Theorem states that there exists a strategy profile σ^{SP} which implements the same conditional approval probabilities under type signal X. Hence, σ^{SP} attains this maximum, which gives V^{SP}.

When $x_f \geq x_f^*(x_s)$, setting $\sigma = (0, 1)$ clearly achieves V^{SP}. When $x_f < x_f^*(x_s)$, we need:

$$\frac{1 - \tilde{\mu_1}}{\tilde{\mu_0}} = \frac{1 - \sigma_f(1 - \mu_1) - \sigma_s\mu_1}{(1 - \sigma_f)\mu_0 + (1 - \sigma_s)(1 - \mu_0)} = \frac{x_f^*(x_s)}{1 - x_f^*(x_s)}$$
$$\frac{\tilde{\mu_1}}{1 - \tilde{\mu_0}} = \frac{\sigma_f(1 - \mu_1) + \sigma_s\mu_1}{1 - (1 - \sigma_f)\mu_0 - (1 - \sigma_s)(1 - \mu_0)} = \frac{x_s}{1 - x_s}$$

One can verify easily that the pair:

$$\sigma_f = 0 \qquad \sigma_s = \frac{1 - 2x_f^*(x_s)}{\mu_1(1 - x_f^*(x_s)) - (1 - \mu_0)x_f^*(x_s)} < 1$$

sovles this system. Since this is a linearly independent system of two equations in two unknowns, the solution is unique.

□

The main insight into Theorem 2 is supplied by Blackwell's Theorem. With the condition that they be symmetric, the planner can choose how evaluators respond to *Success* and *Fail* signals from X. Through Blackwell's Theorem, garbling actions evaluators take in response to information from X can be linked to garbling the information from X itself. In fact, this helps

reformulate the planner's problem as one of garbling the type signal X while obliging them to follow the recommendations from this garbled signal; thus fixing $\sigma = (0, 1)$.

The remainder of the reasoning follows Proposition 3.10. Suppose the planner could weaken the good and bad news that evaluators' type signal X delivers. Weakening good news would never be optimal as $\pi(x_f, x_s)$ is decreasing in x_s for any x_f. Weakening bad news could be optimal for the planner. He would do this if and only if $x_f < x_f^*(x_s)$. Then, his ideal strength for bad news would be $x_f^*(x_s)$. Evaluators' payoffs would then equal $\pi^*(x_f^*(x_s), x_s)$; the highest payoff they can achieve in equilibrium under this strength of good news.

Theorem 2 establishes that this is achieved when evaluators mix between approving and rejecting applicants with *Success* signals from X. All *Fail* applicants are still rejected. Evaluators start mixing once x_f is below $x_f^*(x_s)$. This ensures that a single *Success* signal does not guarantee approval for candidates. Due to the strength of bad news, evaluators are better off when some candidates must get multiple *Success* signals before their eventual approval.

Evaluators mix between approving and rejecting *Success* applicants in some equilibria of the private K-evaluation game, too. They do so when $\pi(x_f, x_s) < 0$; so that approving every *Success* applicant would not be tenable in equilibrium. This strategy alleviates the extent of adverse selection in the pool, and sustains an equilibrium where evaluators are indifferent when approving upon a *Success* signal. The optimal symmetric strategy profile instructs evaluators to mix much earlier: once equilibrium payoffs fall below $\pi^*(x_f, x_s)$. It instructs *Success* applicants to be rejected more frequently, too; to alleviate adverse selection further.

Under this optimal symmetric strategy, no improvement in the type signal decreases evaluators' payoffs. If x_f falls below $x_f^*(x_s)$, evaluators still expect a payoff $\pi^*(x_f^*(x_s), x_s)$. However, strengthening bad news beyond $x_f^*(x_s)$ can never increase their payoffs either, due to the same reason.

Theorem 2 reveals the extent of equilibrium inefficiency in the private K-funding model. Unless history is irrelevant under the type signal X, the optimal symmetric strategy profile affords evaluators higher payoffs than those they can achieve in equilibrium. It does so by effectively *garbling* evaluators' type signal X, and weakening the bad news evaluators' type signals deliver. Evaluators achieve this by mixing between approving and rejecting *Success* applicants. This implies that some candidates must produce multiple *Success* signals before eventually receiving approval.

Next, I consider the optimal *asymmetric* strategy profile. Under this strategy profile, the planner can assign different a different strategy σ_j to every evaluator $j \in \{1, 2, ..., K\}$. Hence, he now solves:

$$\max_{\{\sigma_i\}_{i=1}^K:\ \sigma_i \in [0,1]^2} \sum_{i=1}^{K} \Pi_i(x_f, x_s; \sigma_i, \sigma_{-i}) \tag{FP}$$

I denote the value of this problem as V^{FP}:

$$V^{FP}(x_f, x_s) := \max_{\{\sigma_i\}_{i=1}^K:\ \sigma_i \in [0,1]^2} \left\{ \sum_{i=1}^{K} \Pi_i(x_f, x_s; \sigma_i, \sigma_{-i}) \right\}$$

Theorem 3 describes the optimal *asymmetric* strategy profile.

Theorem 3. *V^{AP} is attained by the strategy profile $\{\sigma_i^{AP}\}_{i=1}^K$, uniquely up to a reordering of evaluators. $\{\sigma_i^{AP}\}_{i=1}^K$ is given by:*

$$\sigma_j^{AP}(f) = 0 \quad \forall\ j \in \{1, 2, ..., K\}$$

$$\sigma_j^{AP}(s) = \begin{cases} 1 & j \leq k^{AP} \\ 0 & j > k^{AP} \end{cases}$$

where:

$$k^{AP} := \min\left\{K, \max\left\{k \in \mathbb{N} : \frac{p_0}{1-p_0}\left(\frac{x_f}{1-x_f}\right)^{k-1}\frac{x_s}{1-x_s} \geq \frac{c}{1-c}\right\}\right\}$$

Under the optimal asymmetric strategy profile, only a subset of evaluators approve any applicants, which I call the *active* evaluators. Active evaluators approve an applicant if and only if she produces a *Success* signal. The *inactive* evaluators reject every applicant they get, irrespective of their signal realisations. The number of *active* evaluators, k^{AP}, is the highest number $\mathcal{K}$ of evaluators such that history would be irrelevant in the a $\mathcal{K}$-evaluation game with type signal (x_f, x_s). Approving a *Success* applicant would be optimal even if she received $\mathcal{K} - 1$ *Fail* signals earlier, but not more.

Proof. Consider the strategy profile $\{\sigma_i\}_{i=1}^K$. The probability that a candidate with type $\theta = L$ is rejected by all evaluators is then:

$$\mathcal{R}^L := \prod_{i=1}^{K}[\mu_0(1-\sigma_i(f)) + (1-\mu_0)(1-\sigma_i(s))]$$

and similarly for a candidate of type $\theta = H$:

$$\mathcal{R}^H := \prod_{i=1}^{K}[(1-\mu_1)(1-\sigma_i(f)) + \mu_1(1-\sigma_i(s))]$$

irrespective of the order in which an applicant applies to evaluators. Evaluators' total payoff, by Proposition 3.1, is given by:

$$\sum_{j=1}^{K}\Pi_j(x_f, x_s, \sigma_i, \sigma_{-i}) = (p_0 - c) + (1-p_0)c\mathcal{R}^L - p_0(1-c)\mathcal{R}^H$$

which again is independent of candidates' order of application to the evaluators. Therefore, we can, without loss of generality, assume that all candidates apply to evaluators in order of evaluators' indices. Then, let:

$$\mathcal{R}_k^L := \prod_{i=k+1}^{K}[\mu_0(1-\sigma_i(f)) + (1-\mu_0)(1-\sigma_i(s))]$$

$$\mathcal{R}_k^H := \prod_{i=k+1}^{K}[(1-\mu_1)(1-\sigma_i(f)) + \mu_1(1-\sigma_i(s))]$$

be the probabilities that a $\theta = L$ and $\theta = H$ candidate, respectively, is rejected by all remaining evaluators, conditional on having received $k < K$ rejections. $\mathcal{R}_K^L = \mathcal{R}_K^H = 0$. Finally, for a given

profile of strategies $\{\sigma_j\}_{j=1}^K$, define the posteriors q_j, $q_j(f)$, and $q_j(s)$ as follows:

$$q_j := \mathbb{P}\left(\theta = H \mid j-1 \text{ rejections given } \{\sigma_i\}_{i=1}^K\right)$$
$$q_j(f) := \mathbb{P}\left(\theta = H \mid j-1 \text{ rejections given } \{\sigma_i\}_{i=1}^K, x_j = x_f\right)$$
$$q_j(s) := \mathbb{P}\left(\theta = H \mid j-1 \text{ rejections given } \{\sigma_i\}_{i=1}^K, x_j = x_s\right)$$

implying:

$$\frac{q_j(f)}{1-q_j(f)} = \frac{q_j}{1-q_j}\frac{x_f}{1-x_f} \qquad \frac{q_j(s)}{1-q_j(s)} = \frac{q_j}{1-q_j}\frac{x_s}{1-x_s}$$

By Bellman's Optimality Principle, the optimal strategy profile $\{\sigma_j^{FP}\}_{j=1}^K$ must then solve:

$$\sigma_j^{AP}(f) \in \arg\max_{\sigma_j(f)\in[0,1]} \sigma_j(f)u(q_j(f)) + (1-\sigma_j(f))V_j(q_j(f)) \tag{3.24}$$
$$\sigma_j^{AP}(s) \in \arg\max_{\sigma_j(s)\in[0,1]} \sigma_j(s)u(q_j(s)) + (1-\sigma_j(s))V_j(q_j(s)) \tag{3.25}$$

where:

$$u(q) = q - c$$

is the flow payoff from accepting an applicant with the posterior $\mathbb{P}(\theta = H) = q$, and:

$$V_j(q) = q\mathcal{R}_j^H(1-c) - (1-q)\mathcal{R}_j^L c$$

is the expected continuation payoff from rejecting the applicant given $\{\sigma_i^{FP}\}_{i=j+1}^K$[10].

Since both 3.24 and 3.25 are linear, $\sigma_j^{SP}(f), \sigma_j^{SP}(s) \in [0,1]$ for all $j \in \{1,2,...,K\}$. First, I show that $\sigma_i^{SP}(f) = 0$ for all $i \in \{1,2,...,K\}$. *Always* rejecting following the kth period is feasible by setting $\sigma_j = 0$ for all $j > k$. Hence, $V_j(q) \geq 0$ for any $j \in \{1,2,...,K\}$ and $q \in [0,1]$.

Now take the first evaluator. We have: $u(q_1(f)) = u(x_f) < 0$, therefore $\sigma_1^{AP}(f) = 0$. Clearly, $\sigma_j^{AP}(f) = 0$ whenever $u(q_j(f)) < 0$ by the same reasoning; it is never optimal to fund a candidate when it is not myopically optimal to do so.

But then, $u(q_j(f)) < 0$ implies $u(q_{j+1}(f)) < 0$; using that $u(q_j(f)) < 0$ implies $\sigma_j^{AP}(f) = 0$:

$$\frac{q_{j+1}(f)}{1-q_{j+1}(f)} = \frac{q_j}{1-q_j}\frac{(1-\mu_1)+\mu_1(1-\sigma_j^{AP}(s))}{\mu_0+(1-\mu_0)\sigma_j^{AP}(s)}\frac{x_f}{1-x_f} \leq \frac{q_j(f)}{1-q_j(f)}$$

Then, $\sigma_i^{AP}(f) = 0$ for all $i \in \{1,2,...,K\}$ and there is some $k^* \in \{1,2,...,K\}$ such that:

$$\sigma_{s,j}^{AP} = \begin{cases} 1 & j \leq k^* \\ 0 & j > k^* \end{cases}$$

That the optimal sequence $\{\sigma_i^{AP}(f)\}_{i=1}^K$ takes a cutoff form is without loss of generality, since the order of the evaluators is irrelevant as pointed out.

I now show that $k^* = k^{AP}$, as proposed in the Theorem. Suppose for contradiction that $k^* > k^{AP}$. But then, an applicant accepted by the $k^{*\text{th}}$ evaluator is accepted after k^*-1 *Fail* signals, and $u(q_{k^*}(s)) < 0$. Suppose otherwise that $k^* < k^{AP}$. Then, $V_k(.) = 0$ for all $k \geq k^*$. However, $u(q_{k^*}(s)) > 0$, yielding a contradiction.

□

[10] Trivially, only when $j < K$.

To mitigate adverse selection into the applicant pool, the optimal asymmetric strategy profile inactivates evaluators until approving an applicant upon a *Success* signal is optimal regardless of his earlier rejections. This strategy profile adjusts the number of evaluators so that history would be irrelevant if only the active evaluators were present. Hence, any improvement in the type signal increases the sum of evaluators' payoffs.

4 History Signals in the K-Evaluation Game

Rather than an uninformative history signal, suppose now that evaluators have a *fully informative* history signal Z_F, which fully reveals every applicant's age:

$$\mathbb{P}(z_F = t \mid \tau = t) = 1$$

I call this the *public* K-evaluation game. Since approved candidates immediately retire, Z_F reveals the past verdicts of every evaluator the candidate applied to earlier. Hence, evaluators' learning dynamics mirror the canonical observational learning model, by Bikhchandani et al., 1992 (BHW).

In the unique equilibrium of the public K-evaluation game, *Fail* signals lead to certain rejections due to the individual obedience constraint O-F. Until a candidate's rejections exceed a critical threshold, evaluators approve upon *Success* signals. This threshold is exceeded when a candidate receives sufficient consecutive *Fail* signals. Past this threshold, a *Success* signal no longer suffices to overturn the unfavourable evidence conveyed by the candidate's rejections. This triggers a *rejection cascade*, following which evaluators reject the candidate irrespective of his type signal.

Recall that in Theorem 3, we defined k^{AP} to be the critical number of *Fail* signals which a single *Success* cannot overturn. The optimal asymmetric strategy profile leaves k^{AP} evaluators active. Likewise, a rejection cascade is triggered following k^{AP} rejections in the equilibrium of the public K-evaluation game. Thus, the equilibrium of the public K-evaluation game mirrors the outcome induced by the optimal asymmetric strategy profile in the private K-evaluation game. In both, a candidate can apply to a maximum of k^{AP} evaluators for approval, who approve him upon a *Success* signal. If they all reject the candidate, every remaining evaluator does so as well.

Evaluators' private information misaggregates in different ways in the equilibria of these games. In the private K-evaluation game, candidates with *too many* Fail signals can get an approval. This owes to evaluators' lack of information about their applicants' age. In the public K-evaluation game, once candidates receive enough initial rejections, they get condemned to rejection thereafter regardless of their future *Success* signals. This owes to the classic herding problem BHW exposed.

Nonetheless, we see that the outcome induced by the *optimal* strategy profile in the private K-evaluation game is equivalent to the *equilibrium* of the public K-evaluation game. This also reveals that with *full information about history*, Z_F, evaluators enjoy higher equilibrium payoffs than they would with *no information about history*, $Z_\emptyset$; regardless of their type signals.

One then wonders whether Z_F affords evaluators a higher equilibrium payoff than any other history signal. In this Section, I first show that this indeed is true; the history signal maximising

evaluators' equilibrium payoffs is the fully informative history signal, regardless of evaluators' type signals. Nonetheless, any improvement in evaluators' history signal does not necessarily raise their equilibrium payoffs. Through a counterexample, I show that there are K-evaluation games where evaluators' equilibrium payoffs decrease when their history signal is improved.

Take a type signal X and history signal Z. Evaluators' S-EQ payoffs in a given period of the steady state are given by:

$$\begin{aligned}\pi^*_Z(x_f, x_s) := (1-c) \times \sum_{t=0}^{K-1} m^*_{H,t}\, \mathbb{E}_{X,Z}\left[\sigma^*(x,z) \mid \theta = H, \tau = t\right] \\ -c \times \sum_{t=0}^{K-1} m^*_{L,t}\, \mathbb{E}_{X,Z}\left[\sigma^*(x,z) \mid \theta = L, \tau = t\right]\end{aligned}$$

where the equilibrium strategies σ^* satisfy condition 2.10 and $m^*_{\theta,\tau}$ are given by equations 2.8 and 2.9. What is the history signal Z which maximises evaluators' equilibrium expected payoffs? Lemma 4.1 establishes that irrespective of the type signal X, Z_F is the optimal history signal.

Lemma 4.1. *Where Z is any history signal and Z_F is the fully informative history signal:*

$$\pi^*_{Z_F}(x_f, x_s) \geq \pi^*_Z(x_f, x_s)$$

for any type signal X.

Proof. Fix a symmetric strategy profile for evaluators, with strategies $\sigma(x,z)$ under type signal X and history signal Z. Given symmetry, the sum of evaluators payoffs does not depend on the application order $\mathcal{P}$ candidates follow. So, keeping $\sigma(x,z)$ fixed, suppose all candidates apply to evaluators in the order of their indices; $\mathcal{P}_\iota(\tau) = \tau + 1$ for every candidate ι.

Then, conditional on the realisation x of the type signal, evaluator j admits her applicant with probabilities:

$$\sigma_j(f) := \mathbb{E}_{Z|\tau=j-1}\left[\sigma(x_f, z)\right] \qquad \sigma_j(s) := \mathbb{E}_{Z|\tau=j-1}\left[\sigma(x_s, z)\right]$$

But then, the total payoffs this yields for evaluators equals the total payoffs they would receive in the private K-evaluation game with the strategy profile $\{\tilde{\sigma}_j\}_{j=1}^K$, where:

$$\tilde{\sigma}_j = (\sigma_j(f), \sigma_j(s))$$

Theorem 3 then shows this sum is maximised when:

$$\tilde{\sigma}_j = \begin{cases} (0,1) & j \leq k^{AP} \\ (0,0) & j > k^{AP} \end{cases}$$

However, the discussion above reveals that $Z = Z_F$ induces this in equilibrium, due to BHW. □

Lemma 4.1 builds on Theorem 3's insight. Theorem 3 showed that a planner who could dictate evaluators' strategies would ensure only candidates receiving a *Success* signal before k^{AP} consecutive *Fail* signals would be approved. Its proof shows that this would be optimal *even*

if the planner could condition an evaluator's strategy on her applicant's age. The public K-evaluation game replicates this outcome in equilibrium. Hence, full information about history maximises applicants' expected payoffs regardless of their type signal X.

The K-evaluation game suffers from a misaggregation of information about quality, θ. Although K IID signals about θ arrive, funding decisions cannot capitalise fully on these. An evaluator observes earlier evaluators' verdicts only noisily, if at all. In addition, these verdicts themselves only reveal earlier evaluators' private information imperfectly.

As evaluators' information about history changes, so does the nature of the misaggregation problem they face. In the public K-evaluation game, candidates with too many *Fail* signals might be approved upon a single *Success* signal. On the other hand, in the private K-evaluation game, projects with many *Success* signals fail to get funding due to initial unlucky draws. In the outset, it is not clear which misaggregation problem would harm evaluators less, and how this would depend on their type signal X. Nonetheless, Lemma 4.1 reveals that full information about history maximises evaluators' payoffs regardless of their type signal X.

Lemma 4.1 might tempt the intuition that an improvement in Z is always beneficial for evaluators' equilibrium payoffs, regardless of their type signal X. Proposition 4.2 establishes that this intuition is misguided.

Proposition 4.2. *There exists some K, a type signal X and history signals $Z^\dagger$, Z, such that $Z^\dagger \succ Z$, and:*

$$\pi^*_{Z^\dagger}(x_f, x_s) < \pi^*_Z(x_f, x_s)$$

I prove Proposition 4.2 by constructing a counter-example. Consider a 3-evaluation game with fresh prior p_0, approval cost c, and type signal (x_f, x_s). Let the type signal be such that history is relevant with any number of evaluators $K \geq 2$:

$$\frac{p_0}{1-p_0}\frac{x_f}{1-x_f}\frac{x_s}{1-x_s} < \frac{c}{1-c}$$

Hence, an evaluator observing a *Success* signal would reject the applicant if she learned that at least one evaluator saw a *Fail* signal before.

Let the history signal Z_λ in this game also be binary, with the possible realisations $z_\lambda \in \{\rho, \alpha\}$. Let its distribution be summarised with the conditional probabilities:

$$\mathbb{P}\left(z_\lambda = \rho \mid \tau < 2\right) = \lambda \qquad\qquad \mathbb{P}\left(z_\lambda = \rho \mid \tau = 2\right) = 1$$

We can interpret the history signal Z_λ as recommending an approval ($z_\lambda = \alpha$) or rejection ($z_\lambda = \rho$) to the evaluator, based on her applicant's age τ. A rejection recommendation tells the evaluator to reject the applicant regardless of her type signal, as the applicant is too old. An approval recommendation tells the evaluator to approve her, provided she sees a *Success*.

The history signal Z_0 perfectly reveals whether the applicant is younger than 2 years old, or not. The history signal Z_1 is completely uninformative about the applicant's age. It is easy to show that the family of signals $\{Z_\lambda\}_{\lambda\in[0,1]}$ are Blackwell ordered, and:

$$Z_{\lambda^\dagger} \succ Z_\lambda \quad \iff \quad \lambda^\dagger < \lambda$$

In the proof I relegated to Section 6, I calibrate the primitives of the model so that evaluators follow signal recommendations in equilibrium under the history signal Z_0; they approve all and only applicants with joint signal realisations (x_s, α):

$$\sigma^*(x_f, \rho) = 0 \qquad \sigma^*(x_f, \alpha) = 0 \tag{4.1}$$

$$\sigma^*(x_s, \rho) = 0 \qquad \sigma^*(x_s, \alpha) = 1 \tag{4.2}$$

In this equilibrium, the sum of evaluators' payoffs mirrors that in the associated 2-evaluation game. Candidates are approved only before their third application, and before that they are approved if and only if evaluators observe a *Success*. However, Theorem 2 showed that in the associated 2-evaluation game, evaluators would be better off if they all rejected with a positive probability upon a *Success*. In fact, raising this probability marginally above zero would increase evaluators' payoffs; it would be equivalent to a marginal reduction in the strength of bad news, without a change in good news.

Suppose now λ is marginally greater than zero. Provided the equilibrium described in equations 4.1 survives, evaluators still reject all candidates aged 2. They also reject younger ones with probability λ, due to the recommendation from Z_λ. This has the effect of raising the rejection probability upon a *Success* in the associated 2-evaluation game. Hence, it increases the sum of evaluators' payoffs from the equilibrium Z_0 induces.

5 Appendix

5.1 Steady State Equilibrium

Take a profile of strategies $\{\sigma_j^*(x,z)\}_{j=1}^K$. Consider applicants of type θ who got τ rejections from evaluators excluding j, before they applied to evalautor j. The mass of these applicants are given by $m_{-j,\tau}^{\theta*}$, where:

$$m_{-j,\tau}^{L*} := \begin{cases} (1-p_0)\frac{K-1}{K} & \tau = 0 \\ m_{-j,\tau}^{L} \times (K-1)^{-1}\,\mathbb{E}_{Z|\tau}\left[\sum_{k\neq j}(1-\mu_0)(1-\sigma_k^*(x_s,z)) + \mu_0(1-\sigma_k^*(x_f,z))\right] & \tau > 0 \end{cases}$$

$$m_{-j,\tau}^{H*} := \begin{cases} p_0\frac{K-1}{K} & \tau = 0 \\ m_{-j,\tau}^{H} \times (K-1)^{-1}\,\mathbb{E}_{Z|\tau}\left[\sum_{k\neq j}\mu_1(1-\sigma_k^*(x_s,z)) + (1-\mu_1)(1-\sigma_k^*(x_f,z))\right] & \tau > 0 \end{cases}$$

Then, evaluator j faces a mass $m_{j,\tau}^{\theta*}$ of applicants of type θ and with τ rejections, where:

$$m_{j,\tau}^{\theta*} = \frac{m_{-j,\tau}^{\theta}}{K-\tau-1}$$

This defines vectors:

$$m_j^{\theta*} = (m_{0,j}^{\theta*}, m_{1,j}^{\theta*}, ..., m_{K-1,j}^{\theta*})$$

for each type $\theta \in \{L, H\}$. The strategy profile $\{\sigma_j^*(x,z)\}_{j=1}^K$ constitutes a *steady state equilibrium* if:

$$\sigma_j^*(x,z) = \begin{cases} 0 & \mathbb{P}\left(\theta = H \mid m_j^{\theta*}, X = x, Z = z\right) - c < 0 \\ \in [0,1] & \mathbb{P}\left(\theta = H \mid m_j^{\theta*}, X = x, Z = z\right) - c = 0 \\ 1 & \mathbb{P}\left(\theta = H \mid m_j^{\theta*}, X = x, Z = z\right) - c > 0 \end{cases} \tag{5.1}$$

5.2 Omitted Results

Proposition 5.1. *For any $\sigma_s \in [0,1]$, the pool prior $\delta(\sigma_s)$ under a given experiment (x_f, x_s) is monotone increasing in x_f.*

The pool prior is not monotone in x_s:

Let $K = 2$. I show that the pool prior does not evolve monotonically with x_s. For the experiment (x_f, x_s), the pool prior $\delta(1)$ is implicitly defined as:

$$\frac{\delta(1)}{1-\delta(1)} = \frac{p_0}{1-p_0}\frac{1+(1-\mu_1(x_f,x_s))}{1+\mu_0(x_f,x_s)}$$

This implies that:

$$\begin{aligned} &\frac{\partial \delta(1)}{\partial x_s} \\ \propto &-\frac{\partial \mu_1}{\partial x_s}(1+\mu_0) - \frac{\partial \mu_0}{\partial x_s}(2-\mu_1) \\ \propto &-\frac{x_f}{1-x_f}(1+\mu_0) + (2-\mu_1) \end{aligned}$$

where I use Lemma 5.3. The final expression approaches $2 - \mu_1 > 0$ as x_f approaches 0 from above, and takes the value $1 - \mu_0 - \mu_1 < 0$ when $x_f = 0.5$. Therefore, its sign changes with x_f.

On the other hand, the K-garbling is Blackwell improved when x_s increases but x_f is held constant; i.e. z_s increases with x_s. Recall from Lemma 5.3 that $\frac{\partial \mu_0}{\partial x_s} > 0$ and $\frac{\partial \mu_1}{\partial x_s} < 0$. Bayes consistency implies:

$$\mathbb{P}(X = x_s)x_s + \mathbb{P}(X = x_f)x_f = 0.5$$

Since x_f is unchanged and x_s is increased, it must be that $\mathbb{P}(X = x_f)$ ($\mathbb{P}(X = x_s)$) increases (decreases) following this change. Since:

$$\frac{z_f}{1-z_f} = \left(\frac{x_f}{1-x_f}\right)^K$$

and $\mathbb{P}(Z = z_f) = \mathbb{P}(X = x_f)^K$, coupled with the corresponding Bayes consistency condition:

$$\mathbb{P}(Z = z_s)z_s + \mathbb{P}(Z = z_f)z_f = 0.5$$

we conclude that z_s must have increased.

Lemma 5.2. *For given (x_f, x_s), equations 2.3 and 2.4 determine (μ_0, μ_1) uniquely.*

Proof. Equations 2.3 and 2.4 can be rewritten as:

$$\begin{bmatrix} x_f & 1-x_f \\ x_s & 1-x_s \end{bmatrix} \times \begin{bmatrix} \mu_0 \\ \mu_1 \end{bmatrix} = \begin{bmatrix} 1-x_f \\ 1-x_s \end{bmatrix}$$

Since

$$\frac{x_s}{1-x_s} > \frac{x_f}{1-x_f}$$

the matrix:

$$M = \begin{bmatrix} x_f & 1-x_f \\ x_s & 1-x_s \end{bmatrix}$$

has full rank, hence is invertible. This establishes our conclusion.

□

Lemma 5.3. *For a binary Blackwell experiment X with induced posteriors (x_f, x_s) and conditional probabilities (μ_0, μ_1):*

$$\frac{\partial \mu_1}{\partial x_s} \Big/ \frac{\partial \mu_0}{\partial x_s} = -\frac{x_f}{1-x_f}$$

and:

$$\frac{\partial \mu_0}{\partial x_s} > 0 \qquad \frac{\partial \mu_1}{\partial x_s} < 0$$

Similarly:

$$\frac{\partial \mu_1}{\partial x_f} \Big/ \frac{\partial \mu_0}{\partial x_f} = -\frac{x_s}{1-x_s}$$

$$\frac{\partial \mu_0}{\partial x_f} > 0 \qquad \frac{\partial \mu_1}{\partial x_f} < 0$$

Proof. From 2.3 and 2.4, we know that:

$$(1-x_s) \times \mu_1 = x_s \times (1-\mu_0) \tag{5.2}$$

$$(1-x_f) \times (1-\mu_1) = x_f \times \mu_0 \tag{5.3}$$

To prove the first part of the claim, differentiate the x_f-*locus* 5.3, which remains intact when x_s changes, with respect to x_s to get:

$$-(1-x_f)\frac{\partial \mu_1}{\partial x_s} = x_f \frac{\partial \mu_0}{\partial x_s}$$

Now, differentiating along the x_s-*locus* 5.2 we get:

$$x_s + (1-x_s)\left[\frac{\partial \mu_1}{\partial x_s} \Big/ \frac{\partial \mu_0}{\partial x_s}\right] = \frac{1-\mu_0+\mu_1}{\partial \mu_0 \,/\, \partial x_s}$$

which, by substitution, implies:

$$x_s - (1-x_s)\frac{x_f}{1-x_f} = \frac{1-\mu_0+\mu_1}{\partial \mu_0 \,/\, \partial x_s}$$

Since the LHS is greater than zero, $\frac{\partial \mu_0}{\partial x_s} > 0$, which implies $\frac{\partial \mu_1}{\partial x_s} < 0$.

The second part of the Lemma is proved similarly.

□

Lemma 5.4. *For any $x_s \in (0.5, 1)$ satisfying O-S, there is some $x_f \in (0, 0.5)$ satisfying O-F such that $\pi^*(x_f, x_s) > 0$.*

Proof. We already know $\pi^*(x_f, x_s) = \max\{0, \pi(x_f, x_s)\}$, so it suffices to show the existence of such pairs (x_f, x_s) where $\pi(x_f, x_s) > 0$. There are two cases to be studied:

- **Case i:** Let $p_0 \leq c$. Then, any $x_f \in (0, 0.5)$ satisfies O-F. $\pi(0.5, x_s) = 0$ for any x_s, and $\frac{\partial \pi(x_f, x_s)}{\partial x_f} |_{x_f=0.5} < 0$. Thus, for some $\varepsilon > 0$, $\pi(0.5 - \varepsilon, x_s) > 0$.

- **Case ii:** Now let $p_0 > c$. Take x_f to be the value solving:

$$\frac{p_0}{1-p_0}\frac{x_f}{1-x_f} = \frac{c}{1-c}$$

Then:

$$\pi(x_f, x_s) \propto \frac{p_0 - c}{(1-p_0)\mu_0^K(1-c)} + \frac{c}{1-c}\left[1 - \left(\frac{x_f}{1-x_f}\right)^{K-1}\right] > 0 \tag{5.4}$$

The continuity of $\pi(x_f, x_s)$ implies that $\exists\ \varepsilon > 0$ such that $\pi(x_f - \varepsilon, x_s) > 0$. Since:

$$\frac{p_0}{1-p_0}\frac{x_f - \varepsilon}{1-(x_f - \varepsilon)} < \frac{c}{1-c}$$

$(x_f - \varepsilon, x_s) \in O_{p_0,c}$.

□

Corollary 5.5. Take a Blackwell experiment X and a strategy pair (σ_f, σ_s). Let $\tilde{\mu_0}$ ($\tilde{\mu_1}$) be the probability that an applicant with type $\theta = L$ ($\theta = H$) gets rejected (accepted) by an evaluator:

$$\tilde{\mu_0} = (1-\sigma_f)\mu_0 + (1-\sigma_s)(1-\mu_0) \qquad \tilde{\mu_1} = \sigma_f(1-\mu_1) + \sigma_s\mu_1 \tag{5.5}$$

Then, there is an experiment X' such that $X \succeq X'$ with the conditional probabilities $(\tilde{\mu_0}, \tilde{\mu_1})$. Conversely, let X' be a Blackwell experiment with conditional probabilities (μ_0', μ_1') and $X \succeq X'$. There exists a strategy pair (σ_f, σ_s) such that 5.5 holds.

Proof. Define the Markov matrices P and M as:

$$P := \begin{bmatrix} \mu_0 & 1-\mu_0 \\ 1-\mu_1 & \mu_1 \end{bmatrix} \qquad M := \begin{bmatrix} 1-\sigma_f & \sigma_f \\ 1-\sigma_s & \sigma_s \end{bmatrix}$$

The diagonal of the Markov matrix PM describes the probability that a $\theta = L$ type is rejected, and a $\theta = H$ type is funded:

$$PM = \begin{bmatrix} (1-\sigma_f)\mu_0 + (1-\sigma_s)(1-\mu_0) & \sigma_f\mu_0 + \sigma_s(1-\mu_0) \\ (1-\sigma_f)(1-\mu_1) + (1-\sigma_s)\mu_1 & \sigma_f(1-\mu_1) + \sigma_s\mu_1 \end{bmatrix}$$

Now let Q be the matrix with the conditional probabilities of experiment X' (the analog of P for X). Blackwell's Theorem states that $X \succ X'$ if and only if there is a Markov matrix M such that $PM = Q$.

□

Proposition 5.6. *No asymmetric equilibrium of the private K-evaluation game can replicate the planner's solution σ^{AP}.*

Proof. First, I show that an asymmetric equilibrium of the private K-evaluation game never exists when $\pi(x_f, x_s) > 0$. When $\pi(x_f, x_s) > 0$, we showed that $\sigma^* = (0, 1)$. $\sigma_f^* = 0$ in any equilibrium. Reducing any σ_s^* below 1 increases the pool prior, hence cannot be an equilibrium.

Therefore an asymmetric equilibrium of the private K-evaluation game might exist only when $\pi(x_f, x_s) \leq 0$. Then, evaluators mix between accepting and rejecting applicants in the symmetric equilibrium.

Let k^{AP} be the number of *active* evaluators the planner would have for experiment X. For there to be an asymmetric equilibrium with k^{AP} active evaluators, a necessary condition is:

$$\frac{p_0}{1-p_0} \frac{\sum_{m=1}^{k^{AP}} (1-\mu_1)^m}{\sum_{m=1}^{k^{AP}} \mu_0^m} \frac{x_s}{1-x_s} \leq \frac{c}{1-c}$$

which can be rewritten as:

$$\frac{p_0}{1-p_0} \frac{1-(1-\mu_1)^{k^{AP}+1}}{1-(\mu_0)^{k^{AP}+1}} \frac{1-\mu_0}{\mu_1} \frac{x_s}{1-x_s} \leq \frac{c}{1-c}$$

Since k^{AP} is the unconstrained planner's optimal solution, we have:

$$\frac{p_0}{1-p_0} \left(\frac{1-\mu_1}{\mu_0}\right)^{k^{AP}-1} \frac{x_s}{1-x_s} \geq \frac{c}{1-c}$$

This implies that:

$$\frac{p_0}{1-p_0} \geq \frac{c}{1-c} \left(\frac{\mu_0}{1-\mu_1}\right)^{k^{AP}-1} \frac{1-x_s}{x_s}$$

Which means that:

$$\frac{p_0}{1-p_0} \left(\frac{1-\mu_1}{\mu_0}\right)^{k^{AP}-1} \frac{x_s}{1-x_s} \geq \left(\frac{\mu_0}{1-\mu_1}\right)^{k^{AP}-1} \frac{\sum_{m=1}^{k^{AP}} (1-\mu_1)^m}{\sum_{m=1}^{k^{AP}} \mu_0^m}$$

□

6 Omitted Proofs

Proof, Proposition 3.1. Take an arbitrary strategy profile $\{\sigma_j\}_{j=1}^K$ and type signal X, which corresponds to experiment (x_f, x_s). Let $\sigma_j = (\sigma_j(x_f), \sigma_j(x_s))$. For both models, we can suppose that a candidate of type θ draws a vector of K signals upon birth:

$$(x_1, x_2, ..., x_K), \quad x_j \in \{x_f, x_s\}$$

where $\mathbb{P}[x_k = x_f \mid \theta = L] = \mu_0$ and $\mathbb{P}[x_k = x_f \mid \theta = H] = 1 - \mu_1$. x_k for $k \in \{1, 2, ..., K\}$ can be thought as the signal the candidate would draw upon her application to the kth evaluator. Suppose that a candidate draws a second K tuple after this:

$$(y_1, y_2, ..., y_K), \quad y_k \in \{0, 1\}$$

where:

$$\mathbb{P}(y_j = 1 \mid x_j = x_f) = \sigma_j(x_f) \qquad \mathbb{P}(y_j = 1 \mid x_j = x_s) = \sigma_j(x_s)$$

y_j can be interpreted as whether or not the applicant would be funded upon applying to evaluator j; $y_j = 1$ ($y_j = 0$) implies he would (not).

Recall that the candidate with index ι applies to evaluator $\mathcal{P}_\iota(\tau+1)$ when he is τ years old, unless he is approved by then; where $\mathcal{P}_\iota \in S_K$. In the simultaneous K-funding model, $\mathcal{P}_\iota$ can be interpreted as applicant ι's tie-breaking order: evaluator $\mathcal{P}_\iota(\tau+1)$'s approval offer is considered by ι if and only if all evaluators $\mathcal{P}_\iota(t)$, $t \leq \tau$, reject him.

Clearly, the probability ι is funded by evaluator j is given by:

$$\mathbb{P}\left(\{y_j = 1\} \cap \left\{y_{j'} = 0 \quad \forall j' \text{ s.t. } \mathcal{P}_\iota^{-1}(j') < \mathcal{P}_\iota^{-1}(j)\right\} \mid \theta_i\right)$$

which is same across both models.

In the simultaneous K-evaluation game, all applicants born in period t are funded (or retire) in the same period. In the K-evaluation game, applicant i born in period t and funded by evaluator j will be funded in period $t + \mathcal{P}_\iota^{-1}(j) - 1$. However, an equal mass of such applicants of types L or H, born in period $t - \mathcal{P}_\iota^{-1}(j) + 1$, are funded in period t. Hence, evaluators' payoffs are equal across models.

□

Proof, Lemma 3.4: Consider *any* equilibrium candidate for the private K-evaluation game: $\{\sigma_j^*\}_{j=1}^K$. This induces a steady state, summarised by masses $m_{\theta,\tau}^{j*}$. The probability that an applicant to evaluator j has type $\theta = H$ is then:

$$\delta^j = \frac{\sum_{\tau=0}^{K-1} m_{H,\tau}^{j*}}{\sum_{\tau=0}^{K-1} m_{L,\tau}^{j*}}$$

$\{\sigma_j^*\}_{j=1}^K$ constitutes an equilibrium if and only if it is optimal for every evaluator, given $\{\delta(j)\}_{j=1}^K$:

$$\sigma_j^*(x_f) \begin{cases} = 0 & \frac{\delta_j^*}{1-\delta_j^*} \times \frac{x_f}{1-x_f} < \frac{c}{1-c} \\ \in [0,1] & \frac{\delta^j}{1-\delta^j} \times \frac{x_f}{1-x_f} = \frac{c}{1-c} \\ = 1 & \frac{\delta^j}{1-\delta^j} \times \frac{x_f}{1-x_f} > \frac{c}{1-c} \end{cases} \qquad \sigma_j^*(x_s) \begin{cases} = 0 & \frac{\delta^j}{1-\delta^j} \times \frac{x_s}{1-x_s} < \frac{c}{1-c} \\ \in [0,1] & \frac{\delta^j}{1-\delta^j} \times \frac{x_s}{1-x_s} = \frac{c}{1-c} \\ = 1 & \frac{\delta^j}{1-\delta^j} \times \frac{x_s}{1-x_s} > \frac{c}{1-c} \end{cases} \tag{6.1}$$

Through this, it is evident that $\sigma_j^*(x_s) \geq \sigma_j^*(x_f)$ for every evaluator j.

However, this implies that $\mathcal{S}(L,\tau;\mathcal{P}) \geq \mathcal{S}(H,\tau;\mathcal{P})$ for any $\mathcal{P} \in S_K$ and $\tau \in \{0,1,...,K-1\}$. This establishes that:

$$\frac{\sum_{\tau=0}^{K-1} m_{H,\tau}^{j*}}{\sum_{\tau=0}^{K-1} m_{L,\tau}^{j*}} \leq \frac{p_0}{1-p_0}$$

pinning down that $\sigma_j^*(x_f) = 0$ for all evaluators j in *any* steady state equilibrium of the private K-evaluation game.

□

Proof, Corollary 3.8. X, described by (x_f, x_s), induces the posterior means x_f and x_s when the evaluator has uniform prior. Under uniform prior, $F(x) := \frac{F^L(x)+F^H(x)}{2}$ is then the distribution of the posterior means X induces. Let X' be the experiment described by (x'_f, x'_s)

Let X be the experiment described below, and X' be described by (x'_f, x'_s), G^L, G^H, and $G(x) := \frac{G^L(x)+G^H(x)}{2}$.

It is well known[11] that X Blackwell dominates X' if and only if F is a mean-preserving spread of G. Define:

$$I_F(x) = \int_0^x F(s)ds \qquad\qquad I_G(x) = \int_0^x G(s)ds \tag{6.2}$$

We demand that:

$$I_F(x) \begin{cases} \geq I_G(x) & x \in [0,1) \\ = I_G(x) & x = 1 \end{cases} \tag{6.3}$$

Then:

$$I_F(x) = \begin{cases} 0 & x \leq x_f \\ \frac{x_s - p_0}{x_s - x_f} \times (x - x_f) & x \in [x_f, x_s] \\ x - p_0 & x \in [x_s, 1] \end{cases} \quad I_G(x) = \begin{cases} 0 & x \leq x'_f \\ \frac{x'_s - p_0}{x'_s - x'_f} \times (x - x'_f) & x \in [x'_f, x'_s] \\ x - p_0 & x \in [x'_s, 1] \end{cases} \tag{6.4}$$

It is clear that $I_F(x_f) \geq I_G(x_f)$ if and only if $x_f \leq x'_f$. Now suppose that $x_s < x'_s$ for contradiction. But then, $I_G(x_s) = \frac{x'_s - p_0}{x'_s - x'_f} \times (x_s - x'_f) > I_F(x_s) = x_s - p_0$ since the function $f(x,y) = (x - y)$ is log-supermodular in (x,y), and $(x'_s, x_s) > (p_0, x'_f)$.

□

Proof, Lemma 3.5: When $\sigma^*_f = 0$, rearrangement yields:

$$\frac{\delta^*}{1-\delta^*} = \frac{p_0}{1-p_0} \times \frac{\sum_{k=0}^{K-1} [\mu_1(1-\sigma^*_s) + (1-\mu_1)]^k}{\sum_{k=0}^{K-1} [\mu_0 + (1-\mu_0)(1-\sigma^*_s)]^k} \tag{6.5}$$

δ^* is monotone in σ^*_s if and only if the Bayes factor $\frac{\sum_{k=0}^{K-1} (1-\mu_1\sigma^*_s)^k}{\sum_{k=0}^{K-1} [1-(1-\mu_0)\sigma^*_s]^k}$ is.

$$\frac{\sum_{k=0}^{K-1} (1-\mu_1\sigma^*_s)^k}{\sum_{k=0}^{K-1} [1-(1-\mu_0)\,\sigma^*_s]^k} = \frac{1-(1-\mu_1\sigma^*_s)^K}{1-(1-(1-\mu_0)\sigma^*_s)^K} \frac{1-\mu_0}{\mu_1} \tag{6.6}$$

therefore, the Bayes factor is monotone decreasing in σ^*_s if and only if $\frac{1-(1-\mu_1\sigma^*_s)^K}{1-(1-(1-\mu_0)\sigma^*_s)^K}$ is. Differentiating this term with respect to σ^*_s and rearranging, it is seen that it has the same sign as:

$$\frac{\mu_1}{1-\mu_0} \left(\frac{1-\sigma^*_s\mu_1}{1-\sigma^*_s(1-\mu_0)} \right)^{K-1} - \frac{1-[1-\sigma^*_s\mu_1]^K}{1-[1-\sigma^*_s(1-\mu_0)]^K} \tag{6.7}$$

[11] See Blackwell, 1953 and Gentzkow and Kamenica, 2016.

Construct the new binary Blackwell experiment $\tilde{X}$ with signal realizations $\{\tilde{x_f}, \tilde{x_s}\}$, such that:

$$\mathbb{P}[\tilde{x} = \tilde{x_s} \mid \theta = H] = \sigma_s^* \mu_1 \qquad \mathbb{P}[\tilde{x} = \tilde{x_f} \mid \theta = H] = 1 - \sigma_s^* \mu_1 \tag{6.8}$$

$$\mathbb{P}[\tilde{x} = \tilde{x_s} \mid \theta = L] = \sigma_s^* (1 - \mu_0) \qquad \mathbb{P}[\tilde{x} = \tilde{x_f} \mid \theta = L] = 1 - \sigma_s^* (1 - \mu_0) \tag{6.9}$$

The associated posterior Bayes factors are:

$$\frac{\mathbb{P}[\tilde{x} = \tilde{x_s} \mid \theta = H]}{\mathbb{P}[\tilde{x} = \tilde{x_s} \mid \theta = L]} = \frac{\mu_1}{1 - \mu_0} > 1 \qquad \frac{\mathbb{P}[\tilde{x} = \tilde{x_f} \mid \theta = H]}{\mathbb{P}[\tilde{x} = \tilde{x_f} \mid \theta = L]} = \frac{1 - \mu_1 \sigma_s^*}{1 - (1 - \mu_0)\sigma_s^*} < 1 \tag{6.10}$$

Now note that the positive term in 6.7 is the Bayes factor for $K - 1$ realizations of $\tilde{x_f}$ and one realization of $\tilde{x_s}$, whereas the negative term is the Bayes factor for *at least* one realization of $\tilde{x_s}$. Therefore, 6.7 is negative for any $K \geq 2$. This establishes that $\frac{d\delta^*}{d\sigma_s^*} < 0$.

The second part of the Lemma now follows from noting that $\frac{\delta^*}{1-\delta^*} = \frac{p_0}{1-p_0}$ when $\sigma_s^* = 0$.

□

Proof, Lemma 3.7: Proposition 3.6 states that $\sigma_f^* = 0$ and $\sigma_s^* \in (0, 1]$. If $\sigma_s^* = 1$, $\pi^*(x_f, x_s) = \pi(x_f, x_s)$. It must also be true that:

$$\frac{\delta^*}{1 - \delta^*} \frac{x_s}{1 - x_s} \geq \frac{c}{1 - c} \tag{6.11}$$

where $\delta^* = \delta(1)$. This implies that an evaluator expects positive payoff from funding a *Success* applicant when others fund every *Success* applicant. Hence, $\pi(x_f, x_s) \geq 0$.

Now suppose $\sigma_s^* < 1$. Then, evaluators' must be indifferent between funding and rejecting a *Success* applicant:

$$\frac{\delta(\sigma_s^*)}{1 - \delta(\sigma_s^*)} \frac{x_s}{1 - x_s} = \frac{c}{1 - c}$$

Therefore their expected equilibrium payoffs are zero. Due to Lemma 3.5:

$$\frac{\delta(1)}{1 - \delta(1)} \frac{x_s}{1 - x_s} < \frac{c}{1 - c}$$

meaning an evaluator would expect a negative payoff from funding a *Success* applicant when others employ $(\sigma_f, \sigma_s) = (0, 1)$; hence $\pi(x_f, x_s) < 0$.

□

Proof, Lemma 3.9: An evaluator's expected payoff from taking her optimal action under posterior $\mathbb{P}(\theta = H) = q$ is given by:

$$V(q) := \max\{0, q - c\}$$

Suppose the type signal X induces posteriors $q_f < p_0 < q_s$ under the evaluator's prior p_0. X satisfies the obedience constraint if:

$$V(q_f) = 0 \qquad V(q_s) = q_s - c$$

It is easy to verify that the expected payoff of the evaluator under type signal X is then given by:

$$(p_0 - q_f) \frac{q_s - c}{q_s - q_f}$$

One can easily verify that:

1. It strictly increases in q_s, and decreases in q_f.

2. Is strictly greater than $\max\{0, p_0 - c\}$, which is the expected payoff from an uninformative signal (one not satisfying the obedience constraint).

□

Proof, Lemma 3.10: Proposition 3.1 established that it suffices to inspect the simultaneous K-evaluation game under strategies $\sigma = (0, 1)$. Suppose that upon her birth, a candidate of type θ draws a vector of K signals:

$$(x_1, x_2, ..., x_K), \quad x_j \in \{x_f, x_s\}$$

where $\mathbb{P}[x_k = x_f \mid \theta = L] = \mu_0$ and $\mathbb{P}[x_k = x_f \mid \theta = H] = 1 - \mu_1$. x_j can be thought as the signal the candidate would draw upon her application to the kth evaluator, with a reasoning that parallels the proof of Proposition 3.1.

In the simultaneous K-evaluation game with $\sigma = (0, 1)$, only projects born in the current period with $x_i = x_f$ for all $i \in \{1, 2, ..., K\}$ would be rejected by *every* evaluator. Evaluators' total payoffs in period t are given by their net gains from rejecting such applicants:

$$(p_0 - c) + (1 - p_0)\mu_0^K c - p_0(1 - \mu_1)^K(1 - c)$$

Since the game is symmetric for every evaluator, a single evaluator's payoff in this case is given by:

$$\pi(x_f, x_s) = K^{-1}\left[(p_0 - c) + (1 - p_0)\mu_0^K c - p_0(1 - \mu_1)^K(1 - c)\right] \tag{6.12}$$

where $\mu_0 = \mu_0(x_f, x_s)$ and $\mu_1 = \mu_1(x_f, x_s)$.

□

Proof, Lemma 3.11: $\pi(x_f, x_s)$ is shown to be:

$$\pi(x_f, x_s) := (p_0 - c) + (1 - p_0) \times \mu_0(x_f, x_s)^K \times c - p_0 \times (1 - \mu_1(x_f, x_s))^K \times (1 - c)$$

in Proposition 3.10. Then:

$$\frac{\partial \pi(x_f, x_s)}{\partial x_s} = K(1 - p_0)\mu_0^{K-1}\frac{\partial \mu_0}{\partial x_s}c + Kp_0(1 - \mu_1)^{K-1}(1 - c)\frac{\partial \mu_1}{\partial x_s}$$

which, by using Lemma 5.3, has the same sign as:

$$\frac{c}{1 - c} - \frac{p_0}{1 - p_0}\left(\frac{x_f}{1 - x_f}\right)^K > 0$$

Thus, $\pi(x_f, x_s)$ is monotone increasing in x_s. Now:

$$\frac{\partial \pi(x_f, x_s)}{\partial x_f} = K(1 - p_0)\mu_0^{K-1}\frac{\partial \mu_0}{\partial x_f}c + Kp_0(1 - \mu_1)^{K-1}(1 - c)\frac{\partial \mu_1}{\partial x_f} \tag{6.13}$$

rearranging and using Lemma 5.3, for fixed x_s:

$$\frac{\partial \pi(x_f, x_s)}{\partial x_f} \propto \frac{c}{1 - c} - \frac{p_0}{1 - p_0}\frac{x_s}{1 - x_s}\left(\frac{x_f}{1 - x_f}\right)^{K-1} \begin{cases} > 0 & x_f < x_f^*(x_s) \\ = 0 & x_f = x_f^*(x_s) \\ < 0 & x_f > x_f^*(x_s) \end{cases} \tag{6.14}$$

where $x_f^*(x_s)$ uniquely solves:

$$\frac{c}{1-c} = \frac{p_0}{1-p_0}\frac{x_s}{1-x_s}\left(\frac{x_f^*(x_s)}{1-x_f^*(x_s)}\right)^{K-1} \tag{6.15}$$

Hence, $\pi(x_f, x_s)$ is strictly quasi-concave and x_f^* is its unique maximiser.

□

Proof, Theorem 1: The argument for the third claim on the sketch proof relies on the fact that $\frac{d\pi(x_f,x_s)}{dx_f}\,|_{x_f=B_f} < 0$ when $B_f \leq x_f^*(x_s)$. This is established in equation 6.14 earlier.

The fourth claim relies on the fact:

$$\pi^*\left(\min\{x_f^*(x_s), B_f\}, x_s\right) = \pi\left(\min\{x_f^*(x_s), B_f\}, x_s\right) > 0$$

I show this in two steps. First, let $B_f = 0.5$, which happens if:

$$\frac{p_0}{1-p_0} < \frac{c}{1_c}$$

Since we cannot have $x_f^*(x_s) = x_s = 0.5$ (the signal is completely uninformative then), we have $x_f^*(x_s) < B_f$. Clearly, $\pi(0.5, x_s) = 0$ (all evaluators always see the uninformative *Fail* signal and reject). But we established in equation 6.14 that $\frac{d\pi(x_f,x_s)}{dx_f}\,|_{x_f=B_f} < 0$ then, which reveals that:

$$\pi\left(x_f^*(x_s), x_s\right) > 0$$

Now let $B_f < 0.5$. This implies that:

$$\frac{p_0}{1-p_0} > \frac{c}{1_c}$$

Now, I will argue through the isomorphic *simultaneous* K-evaluation game that:

$$\pi\left(B_f, x_s\right) > 0$$

Suppose evaluators approved every applicant employing $\sigma = (1, 1)$. Since:

$$\frac{p_0}{1-p_0} > \frac{c}{1_c}$$

evaluators expect to make positive payoffs when every applicant is approved as such. Now, suppose instead that they employ $\sigma = (0, 1)$. Then, only applicants with K consecutive *Fail* signals would be rejected. However, these are the applicants with the lowest posteriors. Hence, evaluators' payoffs cannot have decreased from the strategy profile $\sigma = (1, 1)$.

For the fourth claim, that $\pi(x_f, x_s)$ is a continuous function of both arguments follows from the fact that $\pi(x_f, x_s)$ as a function of μ_0 and μ_1 (equation 3.14) is continuous in both of these arguments, and μ_0 and μ_1 are continuous functions of x_f and x_s.

□

Proof, Proposition 3.17: The auxiliary evaluator observes χ before taking an action. Her expected payoff *before* χ is realised is given by:

$$\pi^A(\xi_f, \xi_s) = \mathbb{E}_\xi\left[\max\{0, \mathbb{E}_{\theta|\xi}\, u^A(\theta, Y)\}\right]$$

It can be established easily that $\mathbb{P}(\theta = H \mid \xi = \xi_s) > \mathbb{P}(\theta = H \mid \xi = \xi_f)$. Also, $X \in \Omega^{p_0,c}$ implies:

$$\frac{p_0}{1-p_0}\frac{\xi_f}{1-\xi_f} = \frac{p_0}{1-p_0}\left(\frac{1-\mu_1}{\mu_0}\right)^K < \frac{p_0}{1-p_0}\frac{1-\mu_1}{\mu_0} = \frac{p_0}{1-p_0}\frac{x_f}{1-x_f} \leq \frac{c}{1-c}$$

Hence, it is never optimal for the auxiliary evaluator to approve after a ξ_f signal. He either rejects every applicant regardless of their signal realisations, or only funds applicants with ξ_s signals. Hence, her expected payoff can be rewritten as:

$$\pi^A(\xi_f, \xi_s) = \max\{0,$$
$$p_0 - c - p_0 \times \mathbb{P}(\chi = \xi_f \mid \theta = H) \times (1-c) + (1-p_0) \times \mathbb{P}(\chi = \xi_f \mid \theta = L)c\}$$

Through 3.20, this makes it immediate that $\pi^A(\xi_f, \xi_s) = \max\{0, K\pi(x_f, x_s)\}$.

□

Proof, Lemma 3.13: First, I handle the case where $\pi^*(x_f, x_s) = 0$. When such:

$$\frac{\delta^*}{1-\delta^*}\frac{x_s}{1-x_s} = \frac{c}{1-c} \tag{6.16}$$

For any $x'_f < x_f$, we have $\pi^*(x'_f, x_s) = 0$. This is because $\pi^*(x_f, x_s) = \max\{0, \pi(x_f, x_s)\}$, and $\pi(x_f, x_s)$ is strictly quasi-concave in x_f, maximised at a strictly positive value. Hence, equation 6.16 continues to hold when $x'_f < x_f$, and δ^* remains unchanged.

Now take the case where $\pi^*(x_f, x_s) > 0$. Then, we have $\sigma^* = (0, 1)$, inducing δ^*. We know, from Lemma 3.5, that reducing σ_s would lead to $\delta > \delta^*$. I will now show the effect of reducing σ_s is equivalent to increasing x_f.

δ is fully pinned down by the probability that a type θ is rejected by the evaluator he applies to. For $\sigma = (0, \sigma_s)$, these probabilities solve the two equations:

$$\frac{\mathbb{P}(H\text{is approved})}{\mathbb{P}(L\text{is approved})} = \frac{\mu_1}{1-\mu_0} \tag{6.17}$$

$$\frac{1-\mathbb{P}(H\text{is approved})}{1-\mathbb{P}(L\text{is approved})} = \frac{(1-\sigma_s)\mu_1 + (1-\mu_1)}{(1-\sigma_s)(1-\mu_0) + \mu_0} \tag{6.18}$$

It is easy to verify that the RHS of 6.18 is a continuously and strictly decreasing function of σ_s, ranging from $\frac{1-\mu_1}{\mu_0} < 1$ to 1.

Under type signal (x_f, x_s) and strategy $\sigma = (0, 1)$, we have:

$$\frac{x_s}{1-x_s} = \frac{\mu_1}{1-\mu_0} = \frac{\mathbb{P}(H\text{is approved})}{\mathbb{P}(L\text{is approved})} \tag{6.19}$$

$$\frac{x_f}{1-x_f} = \frac{1-\mu_1}{\mu_0} = \frac{1-\mathbb{P}(H\text{is approved})}{1-\mathbb{P}(L\text{is approved})} \tag{6.20}$$

As one will observe, the right hand side of equation 6.20 is a continuously and strictly increasing function of x_f, ranging from $\frac{1-\mu_1}{\mu_0}$ to 1 as x_f increases to 0.5. Thus, decreasing σ_s corresponds to increasing x_f, establishing the conclusion through Lemma 3.5.

□

Proof, Proposition 5.1: Since:

$$\begin{aligned}\frac{\delta(\sigma_s)}{1-\delta(\sigma_s)} &= \frac{p_0}{1-p_0} \times \frac{\sum_{k=0}^{K-1}(1-\mu_1\sigma_s)^k}{\sum_{k=0}^{K-1}\left[1-(1-\mu_0)\,\sigma_s\right]^k} \\ &= \frac{p_0}{1-p_0} \times \frac{1-(1-\mu_1\sigma_s)^K}{1-\left[1-(1-\mu_0)\,\sigma_s\right]^K} \times \frac{1-\mu_0}{\mu_1}\end{aligned} \tag{6.21}$$

we have that:

$$\frac{\partial\delta(\sigma_s)}{\partial x_f} \propto \frac{\partial}{\partial x_f}\left[\frac{1-(1-\mu_1\sigma_s)^K}{1-\left[1-(1-\mu_0)\,\sigma_s\right]^K} \times \frac{1-\mu_0}{\mu_1}\right]$$

As $\frac{1-\mu_0}{\mu_1} = \frac{1-x_s}{x_s}$, we can use Lemma 5.3 to manipulate this expression further as:

$$\begin{aligned}&\frac{\partial}{\partial x_f}\left[\frac{1-(1-\mu_1\sigma_s)^K}{1-\left[1-(1-\mu_0)\,\sigma_s\right]^K} \times \frac{1-\mu_0}{\mu_1}\right] \\ \propto\ &\frac{\partial}{\partial x_f}\left[\log\left(1-(1-\sigma_s\mu_1)^K\right) - \log\left(1-(1-\sigma_s(1-\mu_0))^K\right)\right] \\ \propto\ &\frac{(1-\sigma_s\mu_1)^{K-1}\frac{\partial\mu_1}{\partial x_f}}{1-(1-\sigma_s\mu_1)^K} + \frac{(1-\sigma_s(1-\mu_0))^{K-1}\frac{\partial\mu_0}{\partial x_f}}{1-(1-\sigma_s(1-\mu_0))^K} \\ \propto\ &\frac{(1-\sigma_s(1-\mu_0))^{K-1}}{1-(1-\sigma_s(1-\mu_0))^K} - \frac{x_s}{1-x_s}\frac{(1-\sigma_s\mu_1)^{K-1}}{1-(1-\sigma_s\mu_1)^K} \\ \propto\ &\frac{1-(1-\sigma_s\mu_1)^K}{1-(1-\sigma_s(1-\mu_0))^K} - \frac{x_s}{1-x_s}\frac{(1-\sigma_s\mu_1)^{K-1}}{(1-\sigma_s(1-\mu_0))^{K-1}}\end{aligned}$$

Consider again the new experiment $\tilde{X}$ constructed in 6.8. The first term in the last expression above is the Bayes factor for a project which received *at least* one *Success*. The negatively signed term is the Bayes factor for a project which received *just* one *Success*.

□

Proof, Proposition 3.15: One can rearrange:

$$\frac{z_s}{1-z_s} = \frac{1-(1-\mu_1)^K}{1-\mu_0^K}$$

to get:

$$\frac{z_s}{1-z_s} = \frac{\sum_{k=1}^{K-1}(1-\mu_1)^k}{\sum_{k=1}^{K-1}\mu_0^k}\frac{\mu_1}{1-\mu_0} = \frac{1-p_0}{p_0}\frac{\delta(1)}{1-\delta(1)}\frac{x_s}{1-x_s} \tag{6.22}$$

For the first part, recall from Lemma 5.3 that $\frac{\partial\mu_0}{\partial x_s} > 0$ and $\frac{\partial\mu_1}{\partial x_s} < 0$. Bayes consistency implies:

$$\mathbb{P}(X = x_s)x_s + \mathbb{P}(X = x_f)x_f = 0.5$$

Since $x_f^\dagger = x_f$ and $x_s^\dagger > x_s$, it must be that $\mathbb{P}(X^\dagger = x_f^\dagger) > \mathbb{P}(X = x_f)$ and $\mathbb{P}(X^\dagger = x_s^\dagger) < \mathbb{P}(X = x_s)$. Since:

$$\frac{z_f}{1-z_f} = \left(\frac{x_f}{1-x_f}\right)^K \qquad \frac{z_f^\dagger}{1-z_f^\dagger} = \left(\frac{x_f^\dagger}{1-x_f^\dagger}\right)^K$$

and:

$$\mathbb{P}(Z = z_f) = \mathbb{P}(X = x_f)^K \qquad\qquad \mathbb{P}(Z^\dagger = z_f^\dagger) = \mathbb{P}(X^\dagger = x_f^\dagger)^K$$

coupled with the corresponding Bayes consistency conditions:

$$\mathbb{P}(Z = z_s)z_s + \mathbb{P}(Z = z_f)z_f = 0.5 \qquad\qquad \mathbb{P}(Z^\dagger = z_s^\dagger)z_s^\dagger + \mathbb{P}(Z^\dagger = z_f^\dagger)z_f^\dagger = 0.5$$

we conclude that $z_s^\dagger > z_s$ and $z_f^\dagger = z_f$; hence $Z^\dagger \succ Z$.

To see that $x_f^\dagger < x_f$ does not imply $Z^\dagger \succ Z$, let $x_s^\dagger = x_s$. Since $\delta(1)$ is monotone increasing in x_f (Proposition 5.1), through 6.22 we observe that $z_s^\dagger < z_s$. Since $z_f^\dagger > z_f$, these experiments are not Blackwell ranked. □

Proof, Proposition 3.16: To see that $X^\dagger \succ X$ does not imply $Z^\dagger \succ Z$, take $(x_f^\dagger, x_s^\dagger)$ and (x_f, x_s) such that:

$$x_f^\dagger < x_f \qquad\qquad x_s^\dagger = x_s$$

One can rearrange:

$$\frac{z_s}{1 - z_s} = \frac{1 - (1 - \mu_1)^K}{1 - \mu_0^K}$$

to get:

$$\frac{z_s}{1 - z_s} = \frac{\sum_{k=1}^{K-1}(1 - \mu_1)^k}{\sum_{k=1}^{K-1}\mu_0^k}\frac{mu_1}{1 - \mu_0} = \frac{1 - p_0}{p_0}\frac{\delta(1)}{1 - \delta(1)}\frac{x_s}{1 - x_s}$$

Proposition 5.1 established that $\delta(1)$ is increasing in x_f. This implies that z_s is also increasing in x_f. Therefore, $z_f^\dagger > z_f$.

For the second part, recall from Lemma 5.3 that $\frac{\partial \mu_0}{\partial x_s} > 0$ and $\frac{\partial \mu_1}{\partial x_s} < 0$. Bayes consistency implies:

$$\mathbb{P}(X = x_s)x_s + \mathbb{P}(X = x_f)x_f = 0.5$$

Since x_f is unchanged and x_s is increased, it must be that $\mathbb{P}(X = x_f)$ ($\mathbb{P}(X = x_s)$) increases (decreases) following this change. Since:

$$\frac{z_f}{1 - z_f} = \left(\frac{x_f}{1 - x_f}\right)^K$$

and $\mathbb{P}(Z = z_f) = \mathbb{P}(X = x_f)^K$, coupled with the corresponding Bayes consistency condition:

$$\mathbb{P}(Z = z_s)z_s + \mathbb{P}(Z = z_f)z_f = 0.5$$

we conclude that z_s must have increased.

□

Proof, Proposition 4.2: In the counterexample presented in the body, let:

$$K = 3 \qquad p_0 = 0.5 \qquad c = 0.5 \qquad x_f = 0.25 \qquad x_s = 0.7$$

Let $\sigma_\lambda^*(x, z)$ be evaluators' equilibrium strategies under X and Z_λ. Consider the strategy $\tilde{\sigma}_\lambda$ described by:

$$\tilde{\sigma}_\lambda(x, z_\lambda) = \begin{cases} 1 & \text{if } (x, z_\lambda) = (x_s, \alpha) \\ 0 & \text{if } (x, z_\lambda) \neq (x_s, \alpha) \end{cases}$$

A sufficient condition for $\sigma^*_\lambda(x,z) = \tilde{\sigma}_\lambda$ is:

$$\frac{\mathbb{P}(\theta = H \mid z_\lambda = \alpha, \tilde{\sigma}_\lambda)}{\mathbb{P}(\theta = L \mid z_\lambda = \alpha, \tilde{\sigma}_\lambda)} \frac{x_s}{1 - x_s} > \frac{c}{1-c} \tag{6.23}$$

$$\frac{c}{1-c} > \frac{\mathbb{P}(\theta = H \mid z_\lambda = \rho, \tilde{\sigma}_\lambda)}{\mathbb{P}(\theta = L \mid z_\lambda = \rho, \tilde{\sigma}_\lambda)} \frac{x_s}{1 - x_s} \tag{6.24}$$

We have:

$$\mathbb{P}(z_\lambda = \rho \mid \theta, \tilde{\sigma}_\lambda) = \mathbb{P}(\tau = 2 \mid \theta, \tilde{\sigma}_\lambda) + \lambda\, \mathbb{P}(\tau < 2 \mid \theta, \tilde{\sigma}_\lambda)$$
$$\mathbb{P}(z_\lambda = \alpha \mid \theta, \tilde{\sigma}_\lambda) = (1-\lambda)\, \mathbb{P}(\tau < 2 \mid \theta, \tilde{\sigma}_\lambda)$$

and:

$$\mathbb{P}(\tau = 2 \mid \theta = H, \tilde{\sigma}_\lambda) = [(1-\mu_1)(1-\lambda) + \lambda]^2$$
$$\mathbb{P}(\tau = 2 \mid \theta = L, \tilde{\sigma}_\lambda) = [\mu_0(1-\lambda) + \lambda]^2$$

thereby establishing that $\mathbb{P}(\theta = H \mid z_\lambda = \alpha, \tilde{\sigma}_\lambda)$ and $\mathbb{P}(\theta = H \mid z_\lambda = \alpha, \tilde{\sigma}_\lambda)$ are continuous functions of λ.

First, I show that $\sigma^*_0(x,z) = \tilde{\sigma}_0$. Since:

$$\frac{\mathbb{P}(\theta = H \mid z_0 = \rho, \tilde{\sigma}_0)}{\mathbb{P}(\theta = L \mid z_0 = \rho, \tilde{\sigma}_0)} = \left(\frac{p_f}{1-p_f}\right)^2$$
$$\frac{\mathbb{P}(\theta = H \mid z_0 = \alpha, \tilde{\sigma}_0)}{\mathbb{P}(\theta = L \mid z_0 = \alpha, \tilde{\sigma}_0)} = \left(\frac{1-(1-\mu_1)^2}{1-\mu_0^2}\right)$$

then:

$$\frac{\mathbb{P}(\theta = H \mid z_0 = \rho, \tilde{\sigma}_0)}{\mathbb{P}(\theta = L \mid z_0 = \rho, \tilde{\sigma}_0)} \frac{p_s}{1-p_s} < 1$$

and:

$$1 > \left(\frac{\mu_1}{1-\mu_0}\right)^2 \frac{2-\mu_1}{1+\mu_0} > \left(\frac{\mu_1}{1-\mu_0}\right)^2 \frac{1-\mu_1}{\mu_0} = \frac{\mathbb{P}(\theta = H \mid z_0 = \alpha, \tilde{\sigma}_0)}{\mathbb{P}(\theta = L \mid z_0 = \alpha, \tilde{\sigma}_0)} \frac{x_s}{1-x_s}$$

The only remaining task is to show that equations 6.23 and 6.24 hold when λ is marginally increased; so that the equilibrium we posited remains intact. However, this clearly is true since $\mathbb{P}(\theta = H \mid z_\lambda = \alpha, \tilde{\sigma}_\lambda)$ and $\mathbb{P}(\theta = H \mid z_\lambda = \alpha, \tilde{\sigma}_\lambda)$ are continuous functions of λ.

□

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